

# MONETARY POLICY AND SOVEREIGN RISK IN EMERGING ECONOMIES (NK-DEFAULT)\*

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This article develops a New Keynesian model with sovereign default risk. Inflation is set by forward-looking firms, monetary policy is an interest rate rule, and the fiscal government borrows externally, long-term, with an option to default. In this framework, default risk creates inflation pressures through an expectations channel, and tight monetary policy disincentivizes fiscal overborrowing. The model sheds light on temporary inflation events in emerging-market data: short-lived spikes in inflation, spreads, and domestic policy rates. As spreads rise, firms increase their prices in expectation of higher future inflation and low consumption during default. Monetary policy tightens, which reduces inflation and helps bring spreads down by disciplining government borrowing. These monetary-fiscal interactions imply that delivering the flexible-prices allocation may not be optimal for monetary policy. *JEL codes*: F34, F41, E52.

## I. INTRODUCTION

Since the early 2000s, many central banks in emerging markets have successfully reduced inflation to single-digit levels and achieved policy independence from central governments. As in advanced economies, monetary policy in many emerging markets now largely consists of setting domestic nominal interest rates to target inflation. However, the open economy New Keynesian monetary model, a main toolkit for emerging-market central banks, abstracts from sovereign risk, which is a major source of economic

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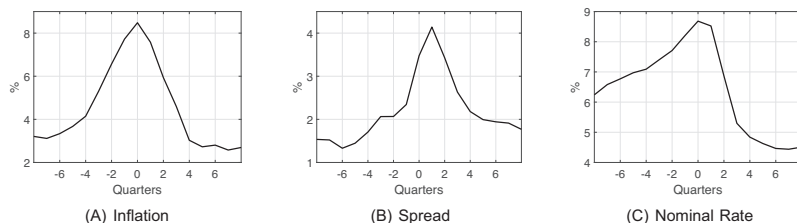


FIGURE I  
Temporary Inflation Events.

fluctuations in these countries. For example, the influential paper by Galí and Monacelli (2005) analyzes monetary policy under perfect financial markets. To bridge this gap, this article presents a New Keynesian framework that integrates sovereign default and investigates the interplay between monetary policy and sovereign risk. Our findings indicate that sovereign risk not only affects real economic activity but also acts as a critical amplifier of inflation, and that monetary policy can discipline sovereign borrowing.

One motivation for our work is the empirical regularity that in emerging markets inflation comoves positively with sovereign risk. We illustrate this pattern in Figure I, which shows the time path for inflation, sovereign spreads, and nominal rates during inflation events, for eight emerging-market inflation targeters between 2004 and 2019.<sup>1</sup> During these events, inflation temporarily rises about 4.5%, sovereign spreads increase by about 2.3%, and central banks increase nominal rates to combat inflation. Notably, the elevated inflation is only temporary, and all variables return to lower levels within approximately a year. We use our model to study these patterns in emerging-market data and perform counterfactual analyses to understand the role of default risk in shaping inflation dynamics and the effect of monetary policy on spreads.

We develop a New Keynesian sovereign-default framework (NK-Default) for a small open economy. Inflation is determined by firms' forward-looking decisions subject to pricing frictions, monetary policy follows an interest rate rule that targets inflation, and

1. The events are constructed with data from Brazil, Chile, Colombia, Mexico, Peru, the Philippines, Poland, and South Africa. Spreads are on dollar-denominated government bonds relative to U.S. bonds. Further information on the sample, variables, and construction of these events is in Section IV.A.

the government chooses whether to repay its debt or default and how much to borrow at prices that compensate for default risk. In this framework, default risk creates inflation pressures and depresses consumption through an expectations channel, and tight monetary policy disincentivizes fiscal overborrowing. High default risk increases expected marginal utility and expected inflation because of the credit restrictions encoded in bond prices and because of productivity losses from default. Through such a change in expectations, a rise in default risk immediately produces higher inflation and lower consumption and output. Monetary policy, in turn, not only affects inflation, consumption, and output but also alters borrowing incentives for the government and therefore default risk. Contractionary monetary policy reduces borrowing incentives because the resulting monetary distortions act as additional costs on government borrowing.

Our integrated NK-Default framework rationalizes emerging-market data, including dynamics and comovements of key macro variables and the temporary inflation events, and sheds light on the drivers of inflation and sovereign spreads. Through counterfactual experiments, we determine that default risk accounts for about half of inflation volatility and show that monetary policy can be designed to reduce spreads substantially. Our framework provides support for the welfare benefits of monetary policy rules in the presence of default risk, across a wide range of robustness exercises.

Our small open economy model consists of households, firms, the monetary authority, and the government. Households value the consumption of foreign and domestic final goods. Consumption is determined through an intertemporal condition that responds to domestic interest rates, set by monetary policy, and to sovereign default risk, which changes expectations of future consumption. Households supply labor to firms which produce varieties of domestic intermediate goods and set their prices under monopolistic competition. The intermediate goods firms are subject to productivity shocks and face frictions in setting their prices, in the tradition of [Rotemberg \(1982\)](#). Final domestic goods firms are competitive and use intermediate goods varieties to produce domestic output, which is consumed by domestic households and exported to the rest of the world. The monetary authority sets a local-currency nominal interest rate, using an interest rate rule subject to monetary policy shocks. Our baseline monetary rule responds to inflation deviations from target, but we also consider alternative rules.

The government borrows internationally with long-term bonds denominated in foreign currency, decides whether to default, and transfers the proceeds from these credit operations to households. Default leads to a temporary exclusion from international financial markets and a reduction in productivity.<sup>2</sup> The price of bonds compensates lenders for default risk, which increases with indebtedness. As in standard New Keynesian models, the interaction of monetary policy and firms' pricing frictions generates monetary distortions, which consist of deviations of inflation from target and inefficient levels of production. In our model, however, default risk also shapes real and nominal variables.

We identify two main monetary-fiscal interactions in our environment. **First, default risk amplifies monetary distortions because it triggers expectations of high inflation and low consumption in the future.** High expected inflation increases current inflation through the pricing decisions of firms, and low expected consumption lowers current consumption through households' intertemporal consumption-smoothing condition, resulting in depressed production. These endogenous costs highlight the significance of default risk for monetary outcomes; we refer to this channel as default amplification. **Second, tight monetary policy can reduce default risk, as monetary distortions tend to disincentivize government borrowing.** The government internalizes the costs of default amplification for the economy and can relax the monetary distortions by reducing default risk. We show that monetary frictions induce wedges in the optimal borrowing condition of the sovereign, enforcing monetary discipline that results in lower borrowing and default risk in equilibrium. Lowering default risk is useful because, without monetary frictions, sovereign-default models with long-term debt exhibit overborrowing and excessively frequent costly defaults due to debt dilution forces: high levels of legacy debt increase borrowing incentives because the resulting decline in bond prices helps dilute the value of existing obligations.

We establish that monetary policy interacts with sovereign risk theoretically in simplified versions of our model and quantitatively in our general model parameterized to emerging

2. Although in our baseline model default leads to a reduction in productivity, we show that our results do not depend on this assumption but it allows for a tighter quantitative fit. In [Section V](#), we explore alternative costs of default, including cases without productivity losses, and find that our conclusions carry through.

-market data. Our theoretical results isolate the default amplification and monetary discipline mechanisms in a tractable setting with preferences that are separable and quasi-linear in foreign-goods consumption. We show that if expectations of future marginal utility and inflation increase with debt, high default risk tends to increase current inflation and reduce output, which is our default amplification mechanism. We then analyze the consequences of monetary policy on default risk by studying a one-time deviation from a constrained-efficient economy. The deviation involves monetary frictions and fiscal overborrowing. We establish our disciplining result: tight monetary policy lowers default risk because the government can counter the increase in the monetary distortion by lowering borrowing and default risk. The disciplining result shows that the standard prescription for monetary policy to neutralize pricing frictions with a strict inflation-targeting regime (which induces the flexible-price allocation) is not optimal in our environment with default risk. In fact, our third theoretical result is that the constrained-efficient outcome can be implemented by a monetary policy rule that targets default risk, as it can neutralize the overborrowing and the pricing frictions.

For our quantitative results, we parameterize the NK-Default model to the data of **eight emerging-market inflation targeters**. Our model produces patterns for spreads, inflation, nominal domestic rates, output, and consumption that resemble these data. **Importantly, the model replicates several untargeted business cycle moments, including the positive comovement of spreads with inflation, nominal rates, and inflation expectations.** We also provide empirical evidence of the disciplining mechanism using panel data. **By projecting sovereign spreads on monetary policy shocks recovered from estimated monetary policy rules, we find that contractionary monetary shocks lower sovereign spreads, consistent with the model's predictions.**<sup>3</sup> We also confront our model with data from the temporary inflation events and find that it can replicate the paths of inflation, output, nominal rates, and spreads with a combination of low productivity shocks and expansionary monetary shocks. The fit of the model over these various dimensions provides support for its mechanisms and credibility for counterfactual analyses.

3. In the estimated interest rate rules, policy rates react positively to inflation, in accordance with the statutory mandate of inflation-targeting central banks.

We perform counterfactual experiments to quantify our amplification and disciplining mechanisms by studying the impact of default risk on inflation outcomes and that of monetary policy on spreads. By comparing with a reference model without default, we find that default risk accounts for about 50% of both the inflation business cycle volatility and its increase during the inflation events. Monetary policy also influences spreads, in addition to other traditional macro outcomes. We find that during the inflation events, a 1% tighter monetary policy could have reduced spreads by about 0.3%. We also consider the effects of monetary policy rules relative to strict inflation targeting. We find that the baseline interest rate rule and one augmented to target default risk reduce mean spreads by 0.5% and 2.2%, respectively, reflecting the disciplining force.

We develop several extensions and show the robustness of our results. We consider the case of local-currency sovereign debt and environments with alternative costs of default: one with loose monetary policy during defaults, one where default does not affect productivity, and an economy with domestic financial frictions shaped by nominal rates and sovereign spreads. We find strong monetary-fiscal interactions in all of these environments, default amplifies inflation volatility and tight monetary policy reduces spreads. When default does not affect productivity, high default risk continues to increase expected marginal utility, a key element for default amplification. Nevertheless, we find that productivity costs help with the quantitative fit of the model, as is known from the sovereign-debt literature.

Our results stress the disciplining properties of monetary policy rules for sovereign debt in environments with monetary commitment, as in the standard New Keynesian models. Using the extension with local-currency debt, we also analyze monetary policy under discretion and confirm the disciplining benefits of rules for sovereign debt. With monetary discretion, policy has an inflationary incentive to depress the real value of debt and, as expected, average inflation doubles. Importantly, discretion also doubles sovereign spreads.

Finally, we assess quantitatively the welfare implications of different monetary policy rules in the baseline model and in the extension economies. We compare welfare in economies under the baseline interest rule and the rule augmented to respond to default risk, relative to the strict inflation-targeting regime. Welfare is higher with our baseline interest rate rule and even higher

under the rule that responds to default risk. Under this default-risk rule, the level of spreads and the volatility of spreads and inflation plummet, which are the sources of the welfare gains. We find similar rankings of monetary policy regimes across all extension economies. Strict inflation targeting is not the optimal monetary regime in economies with default risk. Monetary rules that are responsive enough to inflation or default risk discipline government borrowing and induce less default and higher welfare.

### *I.A. Related Literature*

Our project builds on two distinct strands of the literature on the macroeconomics of emerging markets: quantitative models of sovereign default and New Keynesian open-economy theory. Our government's problem resembles the standard sovereign-default model, in the tradition of [Eaton and Gersovitz \(1981\)](#), as in [Aguiar and Gopinath \(2006\)](#) and [Arellano \(2008\)](#), but with long-term debt, as in [Hatchondo and Martinez \(2009\)](#) and [Chatterjee and Eyigungor \(2012\)](#). We expand this framework to incorporate production, an import–export structure, and pricing frictions. Our domestic monetary environment is close to the workhorse framework of [Galí and Monacelli \(2005\)](#).<sup>4</sup> One methodological difference between our project and standard monetary models is that we use global methods rather than local approximations around the steady state to compute the model.

The literature on sovereign default has recently turned to questions raised by nominal rigidities. Several papers have considered environments with defaultable sovereign debt and downward rigidity of nominal wages. [Na et al. \(2018\)](#) first introduced this friction in a model of sovereign default and emphasized that exchange rate pegs are costly because they prevent devaluations that would adjust real wages to their efficient level. Optimal policy in their environment delivers the joint incidence of devaluations and defaults. [Bianchi, Ottonello, and Presno \(2023\)](#) study the role of downward rigidity of nominal wages for procyclical fiscal policies, which result from a trade-off between fiscal policy stimulating demand and possibly increasing default risk.<sup>5</sup> Our

4. We follow [Galí and Monacelli \(2005\)](#) because they study the small open economy case, in contrast with prior work in a two-country setting, such as [Obstfeld and Rogoff \(1995\)](#) and [Benigno and Benigno \(2003\)](#).

5. Other papers that incorporated downward nominal wage rigidity and sovereign default risk include [Bianchi and Mondragon \(2022\)](#), which studies

project shares the emphasis in these works on the interaction between sovereign risk and nominal rigidities but differs in important ways. First, price frictions in our model arise from optimal, forward-looking price-setting by firms, which results in a standard New Keynesian Phillips curve, where expectations of future inflation matter for current inflation and output. **These papers, in contrast, directly impose that nominal wages are downwardly rigid and abstract from the role of inflation expectations in affecting current inflation and output.** Second, our modeling of monetary policy focuses on a positive theory that resembles the practice of central banks in many emerging markets, which set interest rates to target inflation.<sup>6</sup>

A large literature, following [Calvo \(1988\)](#), studies the incentives of governments to reduce the real value of debt denominated in local currency, by engineering higher-than-expected inflation. [Aguiar et al. \(2013\)](#) analyze the trade-offs generated by monetary policy credibility in a model of self-fulfilling default crises and show that credibility helps suppress self-fulfilling debt crises but hinders the benefits of state-contingent payments induced by inflation.<sup>7</sup> [Hur, Kondo, and Perri \(2018\)](#), [Galli \(2020\)](#), and [Hurtado, Nuño, and Thomas \(2023\)](#) build models with defaultable local-currency debt and a discretionary choice of inflation, whereas [Du, Pflueger, and Schreger \(2020\)](#), [Sunder-Plassmann \(2020\)](#), and [Engel and Park \(2022\)](#) analyze how default and inflation incentives shape the composition of sovereign debt between local and foreign currency. In contrast to these papers, our baseline model focuses on the joint dynamics of endogenous inflation and sovereign risk, with a monetary authority that follows an interest rate rule and a sovereign who issues foreign-currency debt, thus abstracting from the incentive to inflate away the debt. We revisit the case of local-currency debt in one of the extensions of

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rollover crises; [De Ferra and Romei \(2023\)](#) on monetary unions; [Bianchi and Sosa-Padilla \(2024\)](#), which considers the stabilizing role of reserves; and [Roldán \(2025\)](#), which emphasizes precautionary private sector behavior.

6. For the countries in our sample, the statutory objectives assigned by the legislature to central banks center on controlling inflation and are stated in their legal mandates and public statements of goals.

7. Concerning the multiplicity of equilibria and the role inflation can play in selecting among them, [Corsetti and Dedola \(2016\)](#) focus on unconventional monetary policy, whereas [Bacchetta, Perazzi, and van Wincoop \(2018\)](#) analyze how interest rate rules can be used to prevent the self-fulfilling crises in the environment of [Lorenzoni and Werning \(2019\)](#).

our framework and connect our mechanisms to the more familiar one of inflation diluting the value of the nominal debt. More recently, [Maeng \(2025\)](#) studies the currency composition of defaultable sovereign debt in a setting based on our NK-Default framework. Also related, [Espino et al. \(2025\)](#) study monetary policy as arising from the classic incentive to finance fiscal deficits with seigniorage and its interactions with sovereign default risk also with foreign-currency debt. We view this work as complementary to ours and important, especially for emerging markets that have not been able to achieve central bank independence.

An important result of this article is that monetary policy can discipline the overborrowing incentive of the fiscal government. [Aguiar et al. \(2019\)](#) sharply illustrate that sovereign-default models with long-term debt feature overborrowing, because of the debt dilution suffered by legacy lenders, and that the exclusive use of short-term debt can support the constrained-efficient outcome. In this context, [Hatchondo, Martinez, and Sosa-Padilla \(2016\)](#) show that adding no-dilution covenants to long-term contracts can substantially improve outcomes. In our work, we show that monetary policy can provide an alternative avenue for disciplining. We find that under certain conditions, it is possible to design a monetary policy rule that targets default risk so as to achieve the constrained-efficient outcome.

The idea that monetary policy faces a tension between addressing pricing distortions and frictions in international financial markets, a thread running through this study, is present in several classic New Keynesian open-economy studies. [Corsetti, Dedola, and Leduc \(2010\)](#) summarize that the lack of perfect international risk sharing has spillovers on domestic inflation and leads monetary policy to deviate from strict inflation targeting. With incomplete markets, firms' pricing decisions are shaped by inefficient consumption levels, even absent pricing frictions, and the monetary authority may choose to set policy to improve consumption even at the cost of inflation.<sup>8</sup>

[Aoki, Benigno, and Kiyotaki \(2021\)](#) study financial frictions in the banking sector and find an active role for both monetary policy and macroprudential policies. Related, [Basu et al. \(2023\)](#) consider

8. The analysis in the handbook chapter draws on several primary sources, including [Obstfeld and Rogoff \(1995\)](#), [Clarida, Galí, and Gertler \(2002\)](#), [Corsetti and Pesenti \(2001, 2005\)](#); [Benigno and Benigno \(2003, 2026\)](#), and [Devereux and Engel \(2003\)](#).

the use of capital controls and foreign exchange interventions and find that these are superior to interest rate policies, in response to shocks to external inflows. [Itskhoki and Mukhin \(2023\)](#) study the role of optimal monetary policy in fostering international risk sharing and find instead that under certain conditions, a policy that pegs the exchange rate can address both frictions. We focus on financial frictions arising from sovereign default risk, which shape demand patterns as in previous work, but also affect expectations of inflation. In our context, an interest rate rule that targets default risk may address both the monetary frictions and overborrowing incentives.

Finally, our model's implications for exchange rates and international capital flows raise a natural comparison with the work on capital controls, exchange rates, and financial frictions in small open economies, such as [Farhi and Werning \(2016\)](#), [Fanelli \(2017\)](#), [Devereux, Young, and Yu \(2019\)](#), [Itskhoki and Mukhin \(2021\)](#), [Ottonello \(2021\)](#), and [Bianchi and Lorenzoni \(2022\)](#).

## II. MODEL

We consider a small open economy consisting of households, final-goods producers, intermediate goods producers, a monetary authority, and a government conducting fiscal policy. There are three types of goods: final domestic goods, domestic intermediate goods varieties, and foreign imported goods. The final good is produced using all varieties of differentiated intermediate goods, and it is consumed by both domestic and foreign households. Intermediate goods firms produce using labor and set prices in local currency subject to price-setting frictions.<sup>9</sup>

Foreign demand for domestic goods (export demand) is given by

$$X_t = \left( \frac{P_t^d}{\varepsilon_t P_t^*} \right)^{-\rho} \xi,$$

where  $P_t^d$  is the price of domestic goods in local currency,  $P_t^*$  the price of foreign goods in foreign currency,  $\xi$  the level of overall foreign demand,  $\rho$  the export elasticity, and  $\varepsilon_t$  the nominal exchange rate, domestic-currency units per foreign-currency unit.

9. Our sticky-prices setup is a case of producer-currency pricing, following [Gali and Monacelli \(2005\)](#).

An increase in  $\varepsilon_t$  represents a depreciation of the home currency. We assume that the law of one price holds, so we can write the price of the foreign good in local currency as  $P_t^f = \varepsilon_t P_t^*$ . The terms of trade  $e_t$  equal

$$(1) \quad e_t = \frac{P_t^f}{P_t^d} = \frac{\varepsilon_t P_t^*}{P_t^d}.$$

Hence, the foreign demand for domestic goods is a constant elasticity function of the terms of trade and the level of overall foreign demand  $\xi$ :

$$(2) \quad X_t = e_t^\rho \xi.$$

We normalize the foreign price level  $P_t^*$  to one in all periods, thus abstracting from inflation dynamics abroad.

### II.A. Households

Identical households consume domestic goods  $C_t$ , foreign goods  $C_t^f$ , and supply labor  $N_t$ . Their preferences are given by

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u(C_t, C_t^f, N_t),$$

where  $u(C_t, C_t^f, N_t)$  is the per period utility function and  $\beta$  is the discount factor of households.

Households choose consumption, labor supply, and holdings of domestic bonds  $B_t^d$ , taking prices and policies as given. Domestic bonds are denominated in local currency and can only be traded by domestic households. Households own all domestic firms and receive their profits  $\Psi_t$ . They also earn labor income  $W_t N_t$  and receive government transfers  $T_t$ . Their budget constraint is given by

$$P_t^d C_t + (1 + \tau_f) P_t^f C_t^f + q_t^d B_{t+1}^d \leq W_t N_t + B_t^d + \Psi_t + T_t,$$

where  $q_t^d$  is the nominal price of domestic discount bonds and  $\tau_f$  is a constant tax on imports. We can characterize households' choices with the following optimality conditions:

$$(3) \quad - \frac{u_{N,t}}{u_{C,t}} = w_t,$$

$$(4) \quad \frac{u_{C^f,t}}{u_{C,t}} = (1 + \tau_f) e_t,$$

$$(5) \quad u_{C,t} = i_t \beta \mathbb{E}_t \left[ \frac{u_{C,t+1}}{\pi_{t+1}} \right].$$

The real wage is  $w_t = \frac{W_t}{P_t^d}$ , the gross domestic goods inflation, hereafter inflation, is  $\pi_t = \frac{P_t^d}{P_{t-1}^d}$ , and the gross nominal domestic interest rate is the yield of the discount bond price  $i_t \equiv \frac{1}{q_t^d}$ .

### II.B. Final-Goods Producers

The final good  $Y_t$  is produced using a unit measure of differentiated intermediate goods  $y_{jt}$ ,  $j \in [0, 1]$ , under perfect competition, with  $Y_t = [\int_0^1 y_{jt}^{\frac{\eta-1}{\eta}} dj]^{\frac{\eta}{\eta-1}}$ , where  $\eta$  is the elasticity of substitution between intermediate goods. The problem of the final-goods producers yields the standard demand schedule :

$$(6) \quad y_{jt} = \left( \frac{p_{jt}}{P_t^d} \right)^{-\eta} Y_t,$$

where  $p_{jt}$  is the price of intermediate good  $j$  at time  $t$ . The price of domestic goods  $P_t^d$  is the price index  $P_t^d = [\int_0^1 p_{jt}^{1-\eta} dj]^{\frac{1}{1-\eta}}$ .

### II.C. Intermediate Goods Producers

Each differentiated intermediate good is produced with labor  $n_{jt}$ , using a constant return to scale production function, given productivity  $z_t$ :

$$(7) \quad y_{jt} = z_t n_{jt}.$$

Productivity depends on an aggregate shock,  $\tilde{z}_t$ , and the credit standing of the government in international markets,  $\Theta_t$ , such that  $z_t = z(\tilde{z}_t, \Theta_t)$ . As will be discussed shortly, when credit standing is good, productivity is equal to the shock; when credit standing is bad, productivity is lower.<sup>10</sup>

Intermediate goods firms are monopolistically competitive and set the prices of their products, taking as given the demand schedule (6). These firms face price-setting frictions in that they have to pay a quadratic adjustment cost when they do not increase their prices at the inflation rate  $\bar{\pi}$ , as in [Rotemberg \(1982\)](#). Taking as given the nominal wage  $W_t$  and the final-good price  $P_t^d$ , an

10. This assumption is motivated by the evidence that sovereign defaults result in reductions in output and productivity, as documented in [Bai and Zhang \(2012\)](#), [Mendoza and Yue \(2012\)](#), [Hébert and Schreger \(2017\)](#), and [Arellano, Bai, and Bocola \(2026\)](#). We explore alternative costs of default in [Section V.B](#), including a case without productivity costs and one where costs arise endogenously from domestic financial frictions.

intermediate firm  $j$  chooses labor and its price to maximize the present discounted value of profits,

$$\max_{\{n_{jt}, p_{jt}\}} \mathbb{E}_0 \sum_t Q_{t,0} \left\{ p_{jt} y_{jt} - (1 - \tau) W_t n_{jt} - \frac{\varphi}{2} \left( \frac{p_{jt}}{p_{jt-1}} - \bar{\pi} \right)^2 P_t^d Y_t \right\},$$

subject to the production function (7). Firms discount profits using the stochastic discount factor of households,  $Q_{t,0} = \beta^t \frac{u_{C,t} P_0^d}{u_{C,0} P_t^d}$ , and receive a labor subsidy  $\tau$ . The first-order condition for each firm, after imposing symmetry across all firms ( $p_{jt} = P_t^d$ ), results in

$$(8) \quad (1 - \tau) \frac{w_t}{z_t} = \frac{\eta - 1}{\eta} + \frac{1}{\eta} \left\{ \varphi (\pi_t - \bar{\pi}) \pi_t - \mathbb{E}_t \left[ \beta \frac{Y_{t+1}}{Y_t} \frac{u_{C,t+1}}{u_{C,t}} \varphi (\pi_{t+1} - \bar{\pi}) \pi_{t+1} \right] \right\}.$$

This equation is a standard New Keynesian Phillips curve (NKPC) that relates inflation to a measure of contemporaneous unit cost,  $(1 - \tau) \frac{w_t}{z_t}$ , and expected inflation valued by marginal utility next period.

#### *II.D. Government and External Debt*

The government borrows internationally by issuing long-term bonds denominated in foreign currency, with an option to default. To keep long-term debt tractable, we use the maturity structure used by [Hatchondo and Martinez \(2009\)](#) and [Chatterjee and Eyigungor \(2012\)](#). The bond is a perpetuity that specifies a price  $q_t$  and a quantity  $\ell_t$  such that the government receives  $q_t \ell_t$  units of foreign currency in period  $t$ . In the following period, a fraction  $\delta$  of the debt matures, and the government's debt is the sum of legacy debt that has not matured  $(1 - \delta)B_t$  and the new issuance  $\ell_t$  such that  $B_{t+1} = (1 - \delta)B_t + \ell_t$ . Each unit of outstanding bonds calls for a payment every period of  $r + \delta$ .<sup>11</sup>

The government can default on its debt, and, depending on its default history, it is in good or bad credit standing, which is recorded by  $\Theta_t$ . When the government services its debt,  $D_t = 0$ , credit standing is good  $\Theta_t = 0$ , and the government can borrow and decide on the level of debt next period  $B_{t+1}$ . When the government instead chooses to default,  $D_t = 1$ , it avoids the debt payments but suffers a direct utility cost  $v_t$  and a temporary bad credit

11. Note that with this structure, the default-free bond price is equal to one.

standing,  $\Theta_t = 1$ . The utility costs  $v_t$  are i.i.d. enforcement shocks. While in bad credit standing, the government loses access to financial markets, and the productivity of domestic intermediate goods producers is depressed  $z(\tilde{z}_t, \Theta_t = 1) \leq \tilde{z}_t$ . Good credit standing is regained with probability  $\iota$ , at which point the government reenters financial markets with no outstanding debt.

The government transfers to households the proceeds resulting from its trades in international financial markets,  $T_t$ . Conditional on having a good credit standing and repaying the debt, the government's budget constraint in local currency is

$$(9) \quad T_t + \tau W_t N_t = \varepsilon_t [q_t(B_{t+1} - (1 - \delta)B_t) - (r + \delta)B_t] + \tau_f P_t^f C_t^f,$$

where the net capital inflow from debt operations is multiplied by the nominal exchange rate  $\varepsilon_t$  to convert it to domestic currency. When the government is in bad credit standing, its budget constraint has  $B_t = B_{t+1} = 0$ . The constant tax rates for labor and foreign goods consumption,  $\tau$  and  $\tau_f$ , are set as in standard New Keynesian models to correct the markup in goods markets (Rotemberg and Woodford 1999) and allow for a static optimal tariff on imports in steady state, such that  $1 - \tau = \frac{\eta-1}{\eta}$  and  $1 + \tau_f = \frac{\rho}{\rho-1}$ . Using the definition of the terms of trade, equation (1), the government budget constraint in units of domestic goods is

$$t_t + \tau w_t N_t = e_t [q_t(B_{t+1} - (1 - \delta)B_t) - (r + \delta)B_t] + \tau_f e_t C_t^f.$$

The government's objective is to maximize the present discounted value of the flow utility derived from consumption and labor by households,  $\mathbb{E}_0 \sum_{t=0}^{\infty} \beta_g^t u(C_t, C_t^f, N_t)$ . The government's discount factor  $\beta_g$  can differ, in general, from that of households,  $\beta$ . The government borrows from competitive, risk-neutral international lenders discounting the future at a foreign-currency rate  $r$ . The bond price is such that they break even in expectation, thus receiving compensation for any expected losses from default:

$$q_t = \frac{1}{1+r} \mathbb{E}_t [(1 - D_{t+1})(r + \delta + (1 - \delta)q_{t+1})].$$

In states where the government does not default  $D_{t+1} = 0$ , each unit of the bond pays  $r + \delta$  and the  $1 - \delta$  fraction, that does not mature and remains outstanding next period, has market value  $(1 - \delta)q_{t+1}$ . In states in which the government defaults, the associated payoff for lenders is zero. We define the government spread as the difference between the yield-to-maturity of the bond and

the international rate  $r$ , such that

$$\text{spread}_t = (r + \delta) \left( \frac{1}{q_t} - 1 \right).$$

We let  $\Phi_t = \mathbb{E}_t D_{t+1}$  denote the one-period-ahead probability of default.

### II.E. The Monetary Authority

The monetary authority sets policy according to an interest rate rule. In our baseline inflation-targeting regime, the nominal interest rate rule targets domestic-goods inflation. The nominal domestic rate  $i_t$  depends on a long-run value  $\bar{i}$ , responds to the deviation of inflation from target,  $\pi_t$  relative to  $\bar{\pi}$ , and is subject to monetary shocks  $m_t$

$$(10) \quad i_t = \bar{i} \left( \frac{\pi_t}{\bar{\pi}} \right)^{\alpha_P} m_t,$$

with  $\alpha_P > 1$ .

In the following analysis, we also consider a strict inflation-targeting regime, under which the monetary authority sets the nominal rate to keep inflation on target at all times,  $\pi_t = \bar{\pi}$ , a default risk-targeting regime, under which monetary policy uses a rule that reacts to the default risk  $\Phi_t$  directly, and a case of discretionary monetary policy.

### II.F. Equilibrium

We consider a Markov perfect equilibrium, where the government takes into account how its default and borrowing policies affect the allocation of the domestic private economy and the monetary authority's response, now and in the future. The exogenous states are the productivity and monetary shocks,  $s = \{\tilde{z}, m\}$ , and the enforcement shock,  $v$ . The endogenous states for the private and monetary equilibrium are the level of debt  $B$ , the credit standing  $\Theta$ , and borrowing  $B'$ . Recall that the government chooses borrowing  $B'$  only when it has good credit.

When the government enters the period with good credit standing  $\Theta_{-1} = 0$  and debt  $B$ , it chooses whether to default or repay the debt. The default decision determines the end-of-the-period credit standing  $\Theta$ . If it enters the period with bad credit  $\Theta_{-1} = 1$ , the government draws a random variable  $\Lambda$  following a Bernoulli distribution. With probability  $\iota$ ,  $\Lambda = 1$  and the government regains good credit  $\Theta = 0$  if it chooses not to immediately

default again. The evolution of credit standing is given by

$$(11) \quad H_{\Theta}(s, v, B, \Theta_{-1}) = \begin{cases} 0 & \text{if } (\Theta_{-1} = 0 \text{ and } D = 0) \text{ or} \\ & (\Theta_{-1} = 1 \text{ and } \Lambda = 1 \text{ and } D = 0) \\ 1 & \text{otherwise.} \end{cases}$$

The private and monetary equilibrium, which we label  $\Xi(S)$ , depends on the shocks, the debt  $B$ , the credit standing  $\Theta$ , and the government's borrowing choice  $B'$ , because these variables affect government transfers and productivity in the current period. Let  $S = \{s, B, \Theta, B'\}$  be the end-of-period state. The private and monetary equilibrium also depends on the government's equilibrium policy functions for default  $D' = H_D(s', v', B')$ , borrowing  $B'' = H_B(s', B')$ , and the corresponding credit standing  $\Theta' = H_{\Theta}(s', v', B', \Theta)$ , because of the forward-looking nature of the equilibrium.

**DEFINITION 1 (PRIVATE AND MONETARY EQUILIBRIUM).** Given state  $S$ , the government policy functions for default  $H_D(s', v', B')$ , borrowing  $H_B(s', B')$ , credit standing  $H_{\Theta}(s', v', B', \Theta)$ , and the transfer function  $t(S)$  consistent with the government budget constraint, the symmetric private and monetary equilibrium  $\Xi(S)$  consists of

- Households' policies for domestic-goods consumption  $C(S)$ , foreign-goods consumption  $C^f(S)$ , labor  $N(S)$ , and domestic debt  $B^d(S)$ ;
- Intermediate and final goods firms' policies for employment  $n(S)$ , inflation  $\pi(S)$ , final domestic-goods output  $Y(S)$ , and exports  $X(S)$ ; and
- The wage  $w(S)$ , nominal domestic rate  $i(S)$ , and the terms of trade  $e(S)$

such that (i) the policies for households satisfy their budget constraint and optimality conditions (3), (4), (5); (ii) the policies of intermediate and final goods firms satisfy their optimization problems (6), (7), and (8); (iii) export demand equation (2) is satisfied; (iv) the nominal domestic rate satisfies the monetary authority's interest rate rule equation (10); and (v) labor, domestic goods, and domestic bond markets clear, and the balance of payments condition is satisfied.

The labor market clears, with firms' labor demand equal to the labor supplied by households  $n = N$ . Domestic bonds are in zero net supply in the economy, reflected in the market-clearing

condition  $B^d = 0$ . The resource constraint for domestic goods requires that domestic final-goods output equals domestic consumption plus exports and the pricing-adjustment costs,

$$(12) \quad C(S) + X(S) + \frac{\varphi}{2}(\pi - \bar{\pi})^2 Y(S) = Y(S),$$

where aggregate output is given by  $Y(S) = z(\tilde{z}, \Theta) N(S)$ .

The balance of payments condition requires that net imports equal net capital inflows, which here equal the government transfer plus the labor subsidy,

$$(13) \quad e(S)C^f(S)(1 + \tau_f) - X(S) = t(S) + \tau w(S)N(S).$$

The presence of price rigidities leads to inefficient use of labor, as monopolistic firms set time-varying markups. We use a monetary wedge to measure these distortions in production, defined as

$$(14) \quad 1 + \text{monetary wedge} \equiv \frac{z(\tilde{z}, \Theta)}{w(S)} = -\frac{z(\tilde{z}, \Theta)u_C(S)}{u_N(S)}.$$

This wedge captures deviations from production efficiency and depends on the dynamics of current and future inflation, as seen in the NKPC [equation \(8\)](#). Production efficiency requires the marginal product of labor,  $z$ , to be equalized to households' marginal rate of substitution between labor and consumption,  $\frac{-u_N}{u_C}$ .

## II.G. Government Recursive Formulation

We use the following recursive formulation for the government's problem. The bond price  $q(s, B')$  compensates lenders for default risk. It depends on the shocks  $s$  and the borrowing level  $B'$ , because these affect future default probabilities. The bond-price schedule that satisfies the break-even condition for lenders depends on the policy functions of the government for default,  $H_D(s', v', B')$ , and borrowing,  $H_B(s', B')$ , as follows:

$$(15) \quad q(s, B') = \frac{1}{1+r} \mathbb{E} \left[ (1 - H_D(s', v', B')) \times (r + \delta + (1 - \delta)q(s', H_B(s', B'))) \right].$$

Let  $V(s, v, B)$  be the value to the government, including the option to default, such that

$$(16) \quad V(s, v, B) = \max_{D \in \{0, 1\}} \{ (1 - D)W(s, B) + D[W^d(s) - v] \},$$

where  $W(s, B)$  is the value under debt repayment and  $W^d(s) - v$  is the value from defaulting, inclusive of the enforcement shock  $v$ .

The value of repaying is

$$W(s, B) = \max_{B'} \{u(C, C^f, N) + \beta_g \mathbb{E}V(s', v', B')\},$$

subject to the private and monetary equilibrium  $\Xi(S)$  with  $S = \{s, B, \Theta = 0, B'\}$  and the bond-price schedule [equation \(15\)](#). The defaulting value  $W^d$  net of the enforcement cost is given by

$$W^d(s) = u(C, C^f, N) + \beta_g \mathbb{E} \left[ \iota V(s', v', B' = 0) + (1 - \iota) W^d(s') \right],$$

subject to the private and monetary equilibrium  $\Xi(S)$  with  $S = \{s, B = 0, \Theta = 1, B' = 0\}$ . Once in default, the government regains access to the international financial markets with probability  $\iota$ , without outstanding debt. The conditions for the private and monetary equilibrium are collected in [Online Appendix A.1](#).

It is helpful to express the default decision of the government as a cutoff rule based on the enforcement shock  $v$ . Given the i.i.d. nature of the enforcement shock, the default decision  $D(s, v, B)$  can be characterized by a cutoff level  $\hat{v}(s, B)$ , at which the repayment value is equal to the default payoff, such that

$$\hat{v}(s, B) = W^d(s) - W(s, B),$$

and the sovereign is indifferent between the two options. Then  $D(s, v, B) = 1$  whenever  $v \leq \hat{v}(s, B)$  and zero otherwise. Let  $F_v$  be the cumulative distribution function of  $v$ , so that the default probability equals  $\Phi(s, B') = \mathbb{E}_{s'|s} F_v(\hat{v}(s', B'))$ .

We can now define the recursive equilibrium of the economy.

**DEFINITION 2 (EQUILIBRIUM).** Given the aggregate state  $\{s, v, B\}$ , a recursive equilibrium consists of government policies for default  $D(s, v, B)$  and borrowing  $B'(s, B)$ , and government value functions  $V(s, v, B)$ ,  $W(s, B)$ , and  $W^d(s)$ , such that

- Taking as given future policy and value functions,  $H_D(s', v', B')$ ,  $H_B(s', B')$ ,  $V(s', v', B')$ ,  $W(s', B')$ , and  $W^d(s')$ , government policies for default and borrowing and value functions solve its optimization problem.
- Government policies and values are consistent with future policies and values.

## II.H. Government Borrowing

We illustrate the forces shaping debt accumulation, by deriving the government's optimality condition for borrowing, assuming that all functions in the government problem are differen-

table.<sup>12</sup> Optimal borrowing satisfies the following Euler equation

$$(17) \quad u_{Cf} \left[ q + \frac{dq}{dB'} (B' - (1 - \delta)B) \right] (1 - \tau_m^X) - \tau_m^C = \beta_g \mathbb{E} (1 - D') u'_{Cf} [r + \delta + (1 - \delta)q'] (1 - \tau_m^{X'}),$$

which relates the marginal utility of foreign-goods consumption across periods to the bond-price schedule  $q(s, B')$ , future default and borrowing decisions  $D'$  and  $B''$ , future bond prices  $q'$ , and the borrowing wedges  $\tau_m^X$  and  $\tau_m^C$ . [Online Appendix A.2](#) contains a detailed derivation of this equation.

The borrowing wedges  $\tau_m^X$  and  $\tau_m^C$  arise exclusively due to monetary frictions and reflect the Lagrange multipliers associated with the [NKPC](#) and [domestic Euler equations](#) in the government's problem. Without pricing frictions, these wedges are zero. These wedges depend on  $B'$  and are present in this Euler equation because the government internalizes the consequences of its borrowing and default decisions on monetary frictions.

[Equation \(17\)](#) reflects three major forces. The first is the incentive to smooth and tilt the time path of foreign-goods consumption. This force is present in standard models without default risk, like [Galí and Monacelli \(2005\)](#), which exhibit the following undistorted international Euler equation :

$$(18) \quad q u_{Cf} = \beta_g \mathbb{E} u'_{Cf} [r + \delta + (1 - \delta)q'].$$

Absent financial frictions, borrowing only smooths the marginal utility of foreign-goods consumption against shocks and achieves the right tilting of consumption over time, given  $q$  and  $\beta_g$ .

The second force affecting borrowing in our model is the endogenous bond-price schedule  $q$  and the presence of legacy long-term debt  $(1 - \delta)B$ , as emphasized by [Aguilar et al. \(2019\)](#). Bond prices decrease with borrowing due to the increased risk of default,  $\frac{\partial q}{\partial B'} \leq 0$ , and a higher legacy debt  $(1 - \delta)B$  incentivizes borrowing because lower prices dilute this debt,  $-\frac{\partial q}{\partial B'} (1 - \delta)B \geq 0$ . Moreover, with long-term debt the government rolls over only a fraction of debt and therefore has less incentive to reduce debt to secure a higher price for the new issuance. Such debt dilution leads to overborrowing, as established by [Hatchondo, Martinez, and Sosa-Padilla \(2016\)](#). A lower discount for the government,  $\beta_g < \beta$ ,

12. We do not require this assumption for the computation of the model, nor do we employ the Euler equation derived in this section for the numerical implementation.

also leads to overborrowing, as discussed by [Aguiar, Amador, and Fourakis \(2020\)](#).<sup>13</sup>

The third force works through the borrowing wedges  $\tau_m^X$  and  $\tau_m^C$ , and it is unique to our model with sovereign risk and monetary frictions. Positive borrowing wedges lower incentives to borrow. These wedges tend to be positive when the expected marginal utility and the valuation of expected inflation are high, because, as shown in the [Online Appendix](#), the wedges increase with the multipliers on the [domestic Euler equation](#) and the NKPC. Government borrowing tends to increase expected marginal utility because servicing higher debt under the credit restrictions implied by the bond-price schedule depresses consumption. Through the Euler equation, this leads to a decrease in current consumption, increasing the monetary wedge. Higher borrowing also increases the valuation of expected inflation because the high inflation occurs in low-productivity states and becomes more heavily weighed with higher marginal utility, given especially unfavorable bond prices in such states. Higher borrowing increases expected inflation also because of a higher probability of a default, which comes with low productivity and higher inflation. In response, the NKPC calls for increases in current inflation. These extra costs from borrowing in the form of worse monetary frictions are reflected in the borrowing wedges. Importantly, borrowing wedges may nevertheless improve allocations in our model because of overborrowing from dilution and lower discount factors. In the next section, we provide a sharper characterization of these interactions between monetary frictions and default risk in a simplified environment.

### III. THEORETICAL CHARACTERIZATIONS

This section characterizes the interactions between monetary frictions and default risk in a tractable version of our framework and presents our two main mechanisms, default amplification and monetary discipline. We show that default risk worsens monetary distortions by increasing inflation and monetary wedges. This occurs through an expectation channel, as high default risk increases the expectations of future inflation and depressed consumption, which feed into current outcomes through the Euler equation and the NKPC. Tight monetary policy, in turn,

13. [Cuadra and Sapriza \(2008\)](#) and [Hatchondo, Martinez, and Sapriza \(2009\)](#) motivate such discounting as arising from political turnover.

can discipline the government's borrowing and lower default risk by affecting monetary frictions. We relegate all proofs to [Online Appendix B](#).

### III.A. Default Amplification

To show why high default risk tends to increase inflation and induce inefficiently low production, we revisit the key equilibrium NKPC pricing condition

$$(19) \quad (\pi - \bar{\pi})\pi = \frac{\eta - 1}{\varphi} \left( -\frac{u_N}{zu_C} - 1 \right) + \frac{\beta}{Yu_C} \mathbb{E} (Y' u'_C (\pi' - \bar{\pi}) \pi'),$$

where primes (') mark future variables. Inflation increases whenever future expected inflation  $\pi'$  or the marginal utility of consumption  $u'_C$  are high, or when the monetary wedge is low (i.e.,  $-\frac{u_N}{zu_C}$  is high).<sup>14</sup> Default risk affects inflation by affecting these three main terms. Recall that during a default event, productivity is reduced, which implies that inflation is high and consumption is low. Hence, if the risk of a default next period is high, expectations for future inflation and the marginal utility of consumption are high, which increase the expectation term on the right side of the NKPC. Such an expectation channel from default risk gives firms incentives to increase their prices now, raising current inflation.

In response to these inflationary pressures, the monetary policy rule calls for higher interest rates and tight monetary policy. These high nominal domestic rates, however, depress consumption through the domestic Euler equation

$$(20) \quad u_C = i\beta \mathbb{E} \frac{u'_C}{\pi'}.$$

In turn, low domestic consumption tends to reduce production and increase the monetary wedge. All in all, high default risk can result in high inflation and monetary wedges, and tight monetary policy. We call this effect default amplification.<sup>15</sup>

To characterize formally the default amplification mechanism, we simplify the model by assuming that preferences are

14. We find that  $u'_C$  and  $Y'$  tend to move in opposite directions but the net effect is dominated by the marginal utility term. Also, the covariance between  $Y'u'_C$  and  $\pi'$  tends to be positive.

15. We thank Luigi Bocola for his insightful discussion, which led to much of this analysis.

quasi-linear in foreign-goods consumption and are given by

$$(21) \quad u(C, C^f, N) = \log C + C^f - \frac{N^{1+\frac{1}{\xi}}}{1 + \frac{1}{\xi}}.$$

Under these preferences, the consumption of foreign goods fully adjusts to accommodate net capital inflows from debt operations. We evaluate the responses of the model to an increase in default risk  $\mathbb{E}_{s'|s} F_v(\hat{v}(s', B'))$  from higher government borrowing  $B'$ . Higher borrowing affects the rest of the private and monetary equilibrium only through its effect on default risk and the expectation channels: high default risk is reflected in the expectation terms in the NKPC [equation \(19\)](#) and the domestic [Euler equation \(20\)](#). Let functions  $F(s, B', \Theta)$  and  $M(s, B', \Theta)$  encode these expectations, with

$$(22) \quad F(s, B', \Theta) = \mathbb{E} [z(\tilde{z}', \Theta') N(S') u_C(S') (\pi(S') - \bar{\pi}) \pi(S')],$$

$$(23) \quad M(s, B', \Theta) = \mathbb{E} \frac{u_C(S')}{\pi(S')},$$

where the future state  $S' = (s', B', H_\Theta(s', v', B', \Theta), H_B(s', B'))$  depends on the future government policies and the evolution of credit standing. We assume that the functions  $F(s, B', \Theta)$  and  $M(s, B', \Theta)$ , which the government takes as given, are differentiable and increase with borrowing, and then analyze how changes in borrowing affect the equilibrium using a first-order Taylor expansion.

**ASSUMPTION 1 (EXPECTATION TERMS).**  $\frac{\partial F(s, B', \Theta)}{\partial B'} \geq 0$  and  $\frac{\partial M(s, B', \Theta)}{\partial B'} \geq 0$  and the parameters satisfy the restriction  $a_0 \geq \left( \frac{\frac{\partial M(s, B', \Theta)}{\partial B'}}{\frac{\partial F(s, B', \Theta)}{\partial B'}} \right)$ , with  $a_0 = \left[ \frac{\varphi}{(\eta-1) \left( 1 + \frac{\alpha_c + \rho(1-\alpha_c)}{\xi} \right) \beta i \bar{\pi}} \right]$ , where  $\alpha_c$  is the share of domestic consumption in output at the approximating point for the Taylor expansion.

These assumptions ensure that expected inflation and the marginal utility of consumption rise with default risk as  $B'$  increases. These properties, which also hold in our full quantitative model, feed through the equilibrium and affect current allocations and prices. The following proposition characterizes, up to first order, the effects on default risk, inflation, the nominal domestic rate, and the monetary wedge from increased borrowing under [Assumption 1](#), preferences given by [equation \(21\)](#), and when inflation is close to target.

PROPOSITION 1 (AMPLIFICATION). Higher borrowing increases default risk, inflation, the nominal domestic rate, and the monetary wedge.

Borrowing more increases default risk, which in turn affects current inflation and the monetary wedge. High default risk increases expectations of future inflation and marginal utility. Given the stance of monetary policy, the NKPC calls for an increase in current inflation and the domestic Euler equation calls for a decline in current consumption, which increases the monetary wedge. The interest rate rule, in turn, increases the nominal domestic rate, and this monetary policy response dampens the increase in inflation but amplifies the increase in the monetary wedge.

This amplification effect of government borrowing on inflation and the monetary wedge in [Proposition 1](#) depends crucially on the presence of default risk. In our simplified setup here, with quasi-linear preferences, if there were no default risk, higher borrowing would have no effect on inflation, the nominal domestic rate, domestic consumption, or output. In the general model, however, with concave and non-separable preferences over domestic and imported consumption, government borrowing directly affects future consumption, even conditional on repayment. Default risk generates endogenous borrowing limits and an elastic bond-price schedule that further restricts future borrowing capacity. Higher current borrowing therefore reduces future consumption due to these tighter borrowing constraints, causing the expected marginal utility to rise, which makes functions  $M$  and  $F$  increase in  $B'$ . Next we examine how monetary policy interacts with the government's borrowing incentives before presenting these results quantitatively.

### *III.B. Monetary Discipline*

In our framework, not only does default risk amplify monetary frictions, but monetary frictions in turn affect government borrowing and default choices. In particular, tight monetary policy reduces the government's borrowing incentives and lowers default risk. We call this mechanism monetary discipline. We further simplify the model to characterize this mechanism precisely. We alter our environment such that our main frictions—namely, pricing frictions and overborrowing incentives—are only present in period 0, while from period 1 onward the economy's equilibrium

is constrained efficient. We analyze the effects of different monetary policies in period 0 on monetary wedges and default risk, show that strict inflation targeting will not be optimal, and provide an alternative monetary policy rule that can approach arbitrarily close to the constrained-efficient outcome.

To set up the constrained-efficient economy for periods  $t \geq 1$ , we restrict attention to the case of short-term debt  $\delta = 1$  and a government as patient as households,  $\beta_g = \beta$ . The monetary authority eliminates any pricing frictions by setting nominal rates to deliver stable inflation at  $\bar{\pi}$  for periods  $t \geq 1$ . In contrast, in period 0 we assume that  $\beta_g < \beta$ , which induces overborrowing, and monetary policy is given by a set nominal rate  $i$ . We abstract from productivity shocks,  $z_t = \bar{z}$  for any  $t$ , although default continues to reduce productivity,  $z_d \leq \bar{z}$ ; for tractability, we assume the exclusion from international financial markets caused by default is permanent. Default involves an enforcement shock  $\nu$  with a cumulative distribution function  $F_\nu$  and hazard function  $h(\nu)$ , which we assume is strictly increasing. These assumptions allow us to analyze the effects of a one-time deviation from a constrained-efficient environment. The following assumption summarizes the settings for this case.

**ASSUMPTION 2 (ONE-TIME DEVIATION).** For  $t \geq 1$ , the fiscal government is as patient as the household ( $\beta_g = \beta$ ) and the monetary authority implements strict inflation targeting,  $\pi_t = \bar{\pi}$ . In period 0, the government is less patient than households ( $\beta_g < \beta$ ) and the monetary authority sets the domestic nominal rate to a level  $i$ . Absent default, productivity is constant ( $z_t = \bar{z}$ ). Debt is short-term ( $\delta = 1$ ), financial market exclusion is permanent ( $\iota = 0$ ), the hazard function of the enforcement shock  $h(\nu)$  is increasing in  $\nu$ , and preferences are given by [equation \(21\)](#).

Before analyzing policy options during the one-time deviation, we characterize the problem from  $t \geq 1$  onward. In this reference model, pricing frictions are neutralized, and the government borrows and defaults in a constrained-efficient manner. For this analysis, we rely on [Aguilar et al. \(2019\)](#), which characterizes constrained-efficient borrowing in a sovereign-default model and shows that it can be implemented with standard, defaultable short-term bonds. We therefore label this reference economy constrained efficient.

1. *The Constrained-Efficient Economy.* In this economy, the consumption-labor choices are undistorted, and monetary wedges are zero in all states. This implies that inflation is always at target and pricing frictions are neutralized. Furthermore, debt, borrowing, and default do not affect the optimal allocations for domestic consumption, labor, and the terms of trade,  $C^*$ ,  $N^*$ , and  $e^*$ , which satisfy the following three equations,

$$\begin{aligned} C^* + e^{*\rho} &= \bar{z}N^* \\ C^* &= \frac{\rho}{\rho - 1}e^* \\ N^{*\frac{1}{\xi}}C^* &= \bar{z}. \end{aligned}$$

We can substitute these into the utility function and define  $u^* \equiv \log C^* - \frac{(N^*)^{1+\frac{1}{\xi}}}{1+\frac{1}{\xi}} + (e^*)^{\rho-1}$ . The government's problem for the constrained-efficient economy reduces to

$$(24) \quad \begin{aligned} W(B) = \max_{B'} & u^* - (1+r)B + q(B')B' \\ & + \beta \left\{ [1 - F_v(\hat{v}(B'))] W(B') + \int^{\hat{v}(B')} (W^d - v) dF_v(v) \right\}, \end{aligned}$$

subject to the bond price function  $q(B') = 1 - F_v(\hat{v}(B'))$ , and where the default cutoff  $\hat{v}(B')$  satisfies  $\hat{v}(B) = W^d - W(B)$ .

The default value  $W^d$  is the present value of permanent financial autarky  $W^d = \frac{u(C_d, C_d^f, N_d)}{1-\beta}$  subject to the resource constraint  $C_d + e_d^\rho = z_d N_d$  and the balanced-trade condition  $e_d^\rho = e_d C_d^f$ .

The optimal borrowing in the constrained-efficient economy satisfies the following Euler equation:

$$(25) \quad 1 - h(\hat{v}(B^*))B^* = \beta(1+r),$$

and borrowing is at the constrained-efficient level  $B^*$ . Let  $\Phi^* \equiv F_v(\hat{v}(B^*))$  be the equilibrium default risk in the constrained-efficient economy.

To implement the inflation target  $\bar{\pi}$ , the nominal rate in the constrained-efficient economy,  $i^*$ , satisfies the domestic Euler equation,

$$\frac{1}{C^*} = \beta \frac{i^*}{\bar{\pi}} \left[ \frac{1 - \Phi^*}{C^*} + \frac{\Phi^*}{C_d} \right].$$

These constrained-efficient outcomes prevail for all  $t \geq 1$ .<sup>16</sup> These allocations matter for the economy in period 0 because of the expectation terms in the NKPC, the Euler equation, the bond-price function, and the continuation utility.

*2. A One-Time Deviation.* We study a one-time deviation from constrained efficiency. Recall that in period 0 monetary policy is given by the interest rate  $i$  and the government has a lower discount factor  $\beta_g < \beta$ . Inflation may deviate from the target, as called for by the NKPC condition. In period 0, conditional on not defaulting, the government solves the following problem:

$$(26) \quad \max_{B', C, C^f, N} u(C, C^f, N) + \beta_g \left\{ [1 - F_v(\hat{v}(B'))] W(B') + \int^{\hat{v}(B')} (W^d - v) dF_v(v) \right\},$$

subject to the private equilibrium conditions, the bond-price function, and monetary policy  $i$ , which are summarized by:

$$(\text{resource constraint}) \quad C + e^\rho = \left[ 1 - \frac{\varphi}{2}(\pi - \bar{\pi})^2 \right] zN$$

$$(\text{balance payments}) \quad e^\rho = e[C^f + (1 + r)B - q(B')B']$$

$$(\text{intra C-Cf}) \quad C = \frac{\rho}{\rho - 1} e$$

$$(\text{NKPC}) \quad -\frac{u_N}{zu_C} = 1 + \frac{1}{\eta - 1} \varphi(\pi - \bar{\pi})\pi$$

$$(\text{domestic Euler}) \quad \frac{1}{C} = \frac{\beta i}{\pi} \mathbb{E}u_{C'}(B') = \frac{\beta i}{\pi} \left[ \frac{1 - F_v(\hat{v}(B'))}{C^*} + \frac{F_v(\hat{v}(B'))}{C_d} \right]$$

$$(\text{bond price}) \quad q(B') = 1 - F_v(\hat{v}(B')).$$

The expectation terms of this problem are those arising from the constrained-efficient problem. The NKPC condition above reflects the fact that in period  $t = 1$  and onward  $\pi = \bar{\pi}$ ; the expected marginal utility  $\mathbb{E}u_{C'}(B')$  is the weighted average of marginal utility of future consumption without default  $C^*$  and in default  $C_d$ ;

16. See [Online Appendix B.2](#) for a detailed characterization of the constrained-efficient allocations.

the bond-price function and the expected marginal utility use the cutoff function  $\hat{v}$  for an arbitrary state  $B'$ ; and the continuation value function is also that of the constrained-efficient problem for an arbitrary state.<sup>17</sup> Unlike the case of the constrained-efficient economy, the private equilibrium now depends on borrowing and default risk, through the expectations of terms that enter into the domestic Euler equation and bond-price function.

We are now ready to characterize the way in which monetary policy, the nominal rate  $i$ , impacts the equilibrium in the one-time deviation economy. Consider a candidate nominal rate such that the solution to the government problem, [program \(26\)](#), delivers strict inflation targeting  $\bar{\pi}$  in period 0. Denote such a nominal rate in period 0 by  $i^{ST}$ . With strict inflation targeting, the monetary wedge is zero and domestic consumption and labor are efficient,  $(C^{ST} = C^*, N^{ST} = N^*)$ . Optimal borrowing, however,  $B^{ST}$ , is different from the constrained-efficient borrowing since the government has an overborrowing incentive and satisfies

$$(27) \quad 1 - h(\hat{v}(B^{ST}))B^{ST} = \beta_g(1 + r).$$

The government discounts the future more heavily at  $\beta_g < \beta$  and such impatience leads to overborrowing by the government,  $B^{ST} > B^*$ . Overborrowing is costly for households and lowers their welfare level,  $W^{ST}$ . These results are summarized in the next lemma.

**LEMMA 1.** Under [Assumption 2](#), the monetary authority can deliver strict inflation targeting,  $\pi = \bar{\pi}$ , in period 0. If it does so, default risk is higher, and households' welfare is lower than in the constrained-efficient outcome,  $\Phi^{ST} > \Phi^*$ ,  $W^{ST} \leq W^*$ .

The lemma shows that it is possible for the monetary authority to eliminate domestic pricing frictions in period 0 through strict inflation targeting, but that the resulting welfare level is lower than the constrained-efficient case, since the government borrows and defaults too much. The monetary authority, however, can set nominal rates at different levels and affect the economy's outcomes including borrowing and default risk.

Before we explore these effects, it is useful to analyze the effects of default risk on the private economy, particularly on pric-

17. With some abuse of notation, in the resource constraint through bond-price equations, we highlight the Markov aspect of the problem. The future functions are written for arbitrary  $B'$  and are viewed as functions for the government. Of course, all the contemporaneous variables will also be functions of the optimal borrowing choice  $B'$ , but they are chosen as part of the maximization program.

ing frictions. The next lemma shows that given a monetary policy  $i$ , higher default risk increases the monetary wedge.

LEMMA 2. Under [Assumption 2](#), higher default risk  $\Phi$  increases the monetary wedge  $-\frac{zu_C}{u_N}$ , when the monetary wedge is positive.

This result is similar to the amplification result in [Proposition 1](#) but derived for the one-time deviation economy under assumptions on the environment summarized in [Assumption 2](#). Higher default risk increases the future marginal utility of consumption since the expectation in the [domestic Euler equation](#) places more weight on default states, in which consumption is lower,  $C_d \leq C^*$ . In response to this low expected future consumption, current consumption,  $C$ , declines. Low  $C$  also leads to a real appreciation,  $e$  decreases, which in turn reduces exports. Labor is lower because of lower demand for both domestic consumption and exports. As a result, the monetary wedge increases and domestic production is more distorted.

We now characterize equilibrium default risk and monetary wedges for alternative monetary policies in period 0. We find that tighter monetary policy disciplines the fiscal government to borrow less. The following proposition summarizes our disciplining result.

PROPOSITION 2 (DISCIPLINE). Under [Assumption 2](#), if  $i > i^{ST}$ , the monetary wedge is positive and default risk is lower than under strict inflation targeting.

The higher policy rate,  $i > i^{ST}$ , depresses the domestic economy and generates a positive monetary wedge through standard channels in open economy New Keynesian models. It decreases the consumption demand for domestic goods through the [domestic Euler equation](#) and the demand for exports through the resulting exchange rate appreciation, both of which reduce output. The government internalizes the effects of its borrowing on default risk  $\Phi = F_v(\hat{v}(B'))$ , expected marginal utility  $\mathbb{E}u_{C'}$ , domestic consumption  $C$ , and the multiplier on the [domestic Euler equation](#),  $\kappa$ . Its Euler equation for optimal international borrowing reflects such effects, and in this case is

$$(28) \quad 1 - h(\hat{v}(B'))B' - \kappa \frac{1}{1 - F_v(\hat{v}(B'))} \left( \frac{\partial \mathbb{E}u_{C'}(B')}{\partial B'} \frac{u_C}{\mathbb{E}u_{C'}(B')} \right) = \beta_g(1 + r).$$

The government faces an additional cost of borrowing, relative to the case of strict inflation targeting, as  $\kappa \left( \frac{\partial \mathbb{E}u_{C'}(B')}{\partial B'} \frac{u_C}{\mathbb{E}u_{C'}(B')} \right) > 0$  from

**Lemma 2** because increasing  $B'$  increases default risk and the monetary wedge. The additional cost lowers incentives to borrow and therefore reduces default risk.

**Proposition 2** illustrates a trade-off for monetary policy. Strict inflation targeting eliminates pricing frictions, but at the cost of inefficiently high borrowing by the government. Alternatively, a higher policy rate,  $i > i^{ST}$ , induces lower default risk, closer to its efficient level, but at the cost of distorting domestic production. A natural question is whether alternative monetary policy rules could deliver a better outcome or eliminate the trade-off.

We propose an alternative interest rate rule that does deliver better outcomes. The rule targets default risk and dictates increasing nominal rates when default risk rises. We will show that with this rule, which we label the default-risk monetary rule, monetary policy can achieve both efficient default risk and a near-zero monetary wedge. Because of the disciplining effects of high domestic interest rates, this rule discourages excessive borrowing and default risk. We show below that the default-risk monetary rule can be designed to deliver both efficient default risk and efficient production.

**PROPOSITION 3 (DEFAULT-RISK MONETARY RULE).** A monetary policy rule of the form  $i = \bar{i} \Phi^{\alpha^D}$  can achieve the constrained-efficient default risk  $\Phi^*$  and an arbitrarily small monetary wedge of size  $\varepsilon > 0$ , with a positive coefficient  $\alpha^D$ .

Here is the sketch of the proof. Under the default-risk monetary rule, monetary policy responds to the equilibrium default probability  $\Phi = F(\hat{v}(B'))$ , which depends on the fiscal borrowing choices. The government internalizes these effects, giving rise to the following international Euler equation:

$$1 - h(\hat{v}(B'))B' - \kappa \frac{1}{1 - F_v(\hat{v}(B'))} \\ \times \left[ \frac{\partial \mathbb{E}u_C(B')}{\partial B'} \frac{u_C}{\mathbb{E}u_{C'}(B')} + \alpha_{DUC} \frac{f_v(\hat{v}(B'))}{F_v(\hat{v}(B'))} \right] = \beta_g(1 + r).$$

The optimal borrowing condition is modified relative to [condition \(29\)](#) with an additional term,  $\alpha_{DUC} \frac{f_v(\hat{v}(B'))}{F_v(\hat{v}(B'))}$ . This additional term reflects the increased costs of borrowing from tighter monetary policy, which matters through its effects on the multiplier on the domestic Euler equation,  $\kappa$ .

Our approach for implementing the constrained-efficient borrowing is to set  $\alpha_D$  such that the optimal-borrowing condition for the government is the constrained-efficient one, [condition \(25\)](#). This implementation requires setting  $\alpha_D$  to satisfy the following condition:

$$\frac{\kappa}{1 - \Phi^*} \left[ \frac{\partial \mathbb{E} u_{C'}}{\partial B'} \frac{u_C}{\mathbb{E} u_{C'}(B^*)} + \alpha_D u_C \frac{f_v^*}{\Phi^*} \right] = (\beta - \beta_g)(1 + r).$$

The multiplier  $\kappa$  and the marginal utility  $u_C$  in this equation are pinned down by the monetary wedge of size  $\varepsilon$ . The default-risk monetary rule can implement a monetary wedge  $\varepsilon$  by the appropriate choice of  $\bar{i}$  because of its effects through the domestic Euler equation. By setting  $\bar{i} \rightarrow \frac{i^*}{\Phi^{**D}}$  the rule can deliver a monetary wedge of  $\varepsilon \rightarrow 0$ . Inflation, as determined by the NKPC condition, is close to target when  $\varepsilon \rightarrow 0$ . Hence, by targeting only default risk, the optimal monetary rule is able to eliminate two frictions, by achieving both a near-efficient domestic production and constrained-efficient external borrowing.<sup>18</sup>

We conclude this section with a summary of our findings thus far. We have shown that in a simplified version of our model, high default risk can induce high inflation and positive monetary wedges because it affects expectations of future inflation and consumption. Tight monetary policy, in turn, can discipline the government's overborrowing incentives and curb default risk. We also showed that a monetary rule that targets default risk can alleviate the trade-off for monetary policy. In deriving these results, we focused on an economy where default risk affects inflation and monetary wedges only through the expectation channel and abstracted from shocks and preferences that are concave in imported consumption. Next we analyze our general model and show numerically that our key theoretical results are present and shape much of the quantitative outcomes.

#### IV. QUANTITATIVE AND DATA ANALYSIS

We proceed to the quantitative analysis of our model and its mapping to the data. We start by describing key patterns in emerging-market data. We document comovements of inflation, nominal domestic rates, and spreads, discuss properties of tem-

18. We also analyzed rules that target nominal devaluation rates and found that such a policy will not affect borrowing incentives and therefore cannot be designed to achieve constrained-efficient outcomes.

porary inflation events, and provide evidence of the key disciplining mechanism in the model. We describe the parameterization of the model, analyze decision rules and impulse response functions, and compare the model's predictions to the data. We show that our model can fit the data across several untargeted dimensions, including key comovements, time paths, and the elasticities of spread to monetary shocks and that of inflation expectations to default risk. In drawing our lessons, we also compare our baseline NK-Default model to a reference model without default risk and consider alternative monetary policy regimes.

#### *IV.A. Emerging-Market Inflation-Targeters Data*

Several emerging markets successfully adopted inflation targeting as their monetary policy regime in the early 2000s.<sup>19</sup> We analyze data from these countries and document stylized facts about the volatility and comovements of inflation, spreads, and domestic nominal rates. We also compile an event analysis around periods of elevated inflation and provide some evidence for our disciplining and amplification mechanisms.

*1. Stylized Facts.* We collect data on inflation, spreads, nominal domestic rates, and output for eight emerging markets that are inflation targeters. The sample of countries consists of those included in the JP-Morgan Emerging Market Bond Index (EMBI+) that have also successfully adopted inflation targeting. The data start in 2004, by which point all countries considered had adopted the monetary policy regime, and run through 2019.

[Table I](#) reports key statistics on the joint behavior of these series using quarterly data. Inflation is measured by the Consumer Price Index (CPI) and computed as the log difference in the index relative to four quarters prior. Spreads are measured as the difference in yields between foreign-currency government bonds of these emerging markets, as captured by the EMBI index and a U.S. government bond of comparable duration. Domestic nominal rates are the nominal policy rates of central banks, and output is the four-quarter difference in log real GDP. [Online Appendix C.1](#) reports detailed definitions and sources for the data.

We highlight several findings. Inflation is low on average and relatively stable. Mean inflation across these countries ranges be-

19. See [Roger \(2009\)](#) and [Ha, Kose, and Ohnsorge \(2019\)](#) for more details on the implementation and performance of inflation targeting in emerging markets.

TABLE I  
EMERGING-MARKET INFLATION TARGETERS, KEY STATISTICS

|              | Mean      |        | Std. dev. rel. output |        | Correlation with spread |               |        |
|--------------|-----------|--------|-----------------------|--------|-------------------------|---------------|--------|
|              | Inflation | Spread | Inflation             | Spread | Inflation               | Domestic rate | Output |
| Brazil       | 5.6       | 2.8    | 0.6                   | 0.3    | 0.5                     | 0.6           | -0.3   |
| Chile        | 3.2       | 1.4    | 0.8                   | 0.2    | 0.4                     | 0.2           | -0.6   |
| Colombia     | 4.3       | 2.3    | 0.9                   | 0.5    | 0.6                     | 0.4           | -0.3   |
| Mexico       | 4.1       | 2.2    | 0.4                   | 0.3    | 0.2                     | 0.1           | -0.5   |
| Peru         | 2.9       | 2.0    | 0.5                   | 0.3    | 0.5                     | 0.1           | 0.0    |
| Philippines  | 3.9       | 2.1    | 1.4                   | 0.8    | 0.6                     | 0.6           | -0.5   |
| Poland       | 2.1       | 1.1    | 0.9                   | 0.4    | 0.4                     | 0.2           | -0.4   |
| South Africa | 5.0       | 2.2    | 1.2                   | 0.5    | 0.5                     | 0.1           | -0.7   |
| Mean         | 3.9       | 2.0    | 0.8                   | 0.4    | 0.5                     | 0.3           | -0.5   |

*Notes.* Quarterly data, 2004Q1–2019Q4. Inflation is CPI four-quarter log difference. Spread is the JP Morgan EMBI+ Spread. Output is the four-quarter log difference of real GDP. See [Online Appendix C.1](#) for the construction of our sample.

tween 2.1% and 5.6%, with an overall average of 3.9%. The standard deviation of inflation relative to that of output ranges from 0.4 to 1.4, with an average across countries of 0.8. This stable inflation contrasts sharply with the historical experience of these countries, with episodes of very high and volatile inflation. Emerging-markets bond spreads are on average 2% and with a mean standard deviation of 0.9%, which is 0.4 relative to that of output.

We also report correlations of spreads with inflation, nominal domestic rates, and output. Spreads are negatively correlated with output, with an average correlation of  $-0.5$  in this sample. Correlations of spreads with nominal rates and inflation are positive, on average 0.3 and 0.5, respectively. Note that domestic nominal rates are denominated in local currency whereas government spreads are in foreign currency, so that the correlation is not driven by a common domestic inflation or exchange rate factor but instead likely reflects the relationship between inflation and default risk.<sup>20</sup>

20. For a similar time period, we found a wide range of correlations between inflation and spreads for non-inflation targeting emerging countries: the correlation is  $-3\%$  for Argentina,  $-4\%$  for Bulgaria, and  $51\%$  for Ecuador. Therefore, we conclude that the positive correlation between inflation and spreads may not be unique to inflation-targeting emerging countries.

2. *Temporary Inflation Events.* Although the emerging markets in our sample have successfully kept average inflation low, these countries have experienced several events, over the study period, during which inflation rose temporarily by a few percentage points. Importantly, these temporary inflation events have been relatively subdued, and inflation returned to lower levels shortly after the monetary authority tightened policy. We illustrate the dynamics of key variables during these events and rely on these findings for the quantitative analysis of our model.

We classify a country as experiencing an inflation event if inflation is unusually high, defined as two standard deviations or more above its mean. We center the events around the peak in inflation, where the standard deviation is calculated country by country, and analyze the dynamics eight quarters before and after this peak. [Figure II](#) plots average paths across the nine events we found in our data set for inflation dynamics, domestic nominal rates, spreads, and output.<sup>21</sup> The underlying series are standardized country by country; therefore the units in the figure are standard deviations, and a value of zero means the series is at its average. Panel A illustrates the tent-like shape of inflation during the event. Inflation starts slightly below its average level and increases close to 2.5 standard deviations at its peak, before returning to low levels. The 2.5 standard deviation increase corresponds to an average of 4.5 percentage points. Panel B plots the domestic nominal rate, which increases by about 1.2 standard deviations and then falls at the end of the event. At its peak, the domestic nominal rate increase corresponds to 2.9 percentage points above its mean. Output, in Panel C, starts above its average and falls about 2.5 standard deviations, corresponding to a 5.7 percentage point recession, before recovering toward the end of the event. Panel D plots spread dynamics. Spreads also feature a tent-like shape, increasing about 2.5 standard deviations from start to peak, corresponding to about 2.3 percentage points, and falling back to their long-run average by the end of the event.<sup>22</sup>

21. The events have the following peak dates: Brazil 2015Q4, Chile 2008Q3, Colombia 2008Q4 and 2015Q2, Mexico 2008Q4 and 2017Q4, Peru 2008Q4, Philippines 2008Q4, and South Africa 2008Q3.

22. Emerging markets experienced an inflation episode starting in 2021, which featured dynamics similar to those documented here. Inflation increased substantially in all of our eight countries and central banks responded by tightening aggressively, after which inflation fell and broadly returned to target by early

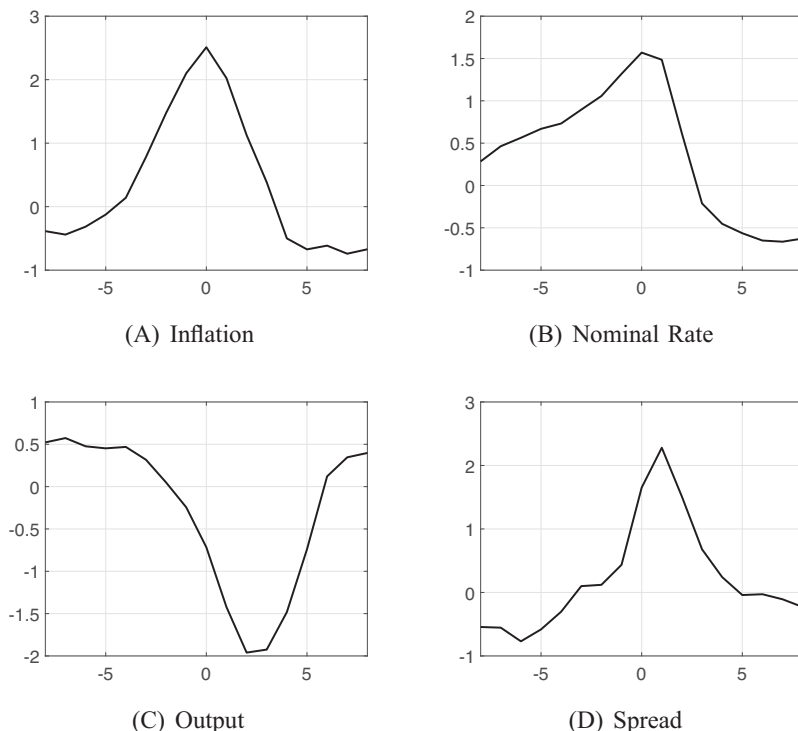


FIGURE II

## Temporary Inflation Events

Average paths around spikes in inflation, in excess of two standard deviations, based on the nine events in our emerging-markets sample. All series are quarterly and standardized country by country so that units on the vertical axes are standard deviations, and zero is the sample average. [Online Appendix C.1](#) describes our sample.

These dynamics illustrate how emerging-market inflation targeters have been successful at managing shocks that lead to temporarily high inflation. The resolution of these inflation episodes is impressive, as inflation decreased and returned to target in about a year with monetary tightening. We also find it noteworthy that spreads are high during these elevated-inflation events. The standard, open economy New Keynesian model of [Galí and Monacelli \(2005\)](#) is silent about sovereign spreads, yet this

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2024. Spreads spiked together with inflation and have since reverted. [Online Appendix C.4](#) contains more details.

empirical regularity in emerging markets is consistent with one of the main mechanisms in our NK-Default framework, that default risk can be an amplifying force for elevated inflation, a mechanism we explored theoretically in [Section III](#). In [Online Appendix C.3](#), we also document that, consistent with our amplifying mechanism, increases in spreads are associated with elevated survey-based inflation expectations for our sample of emerging-market inflation targeters.

*3. Disciplining Effect in the Data.* Using our data sample, we can provide empirical evidence of the disciplining mechanism in our model that tight monetary policy reduces government spreads. As we illustrated already, unconditionally, nominal rates tend to rise with sovereign spreads, as seen by the overall positive correlation and during inflation events. This unconditional positive correlation is of course silent on the direct effect of monetary policy on spreads, which is the object of interest for the disciplining effect.<sup>23</sup> To tease out the direct effect of monetary policy on spreads, we follow the monetary literature and first recover monetary shocks. Then we project government spreads on these estimated monetary shocks.

For the first step, we recover monetary policy shocks by estimating the reaction function of the monetary authority's policy rates to inflation. In particular, for each country  $c$ , we recover the monetary shock  $m_{c,t}$  by projecting policy rates on inflation and lagged policy rates, such that  $i_{c,t} = \alpha_c^0 + \alpha_c^1 i_{c,t-1} + \alpha_c^2 \pi_{c,t} + m_{c,t}$ , where we allow the residuals to be autocorrelated. In estimating these regressions, we follow the literature and estimate them with monthly data. The coefficients  $\alpha_c^2$  are positive for all our countries and statistically significant for six of them. The finding that policy rates tend to react positively to inflation is consistent with the statutory inflation-targeting regime of the countries in our sample.

We estimate the effects of monetary shocks on spreads with a panel regression, where we project spreads on the estimated monetary policy shocks. For our baseline panel regression, we standardize all series at the country level. We project spreads for coun-

23. The unconditional positive correlation of nominal rates and spreads can reflect a third factor affecting both. If, for example, low productivity is associated with increases in default risk and inflation, then a productivity-driven business cycle can rationalize the unconditional correlation.

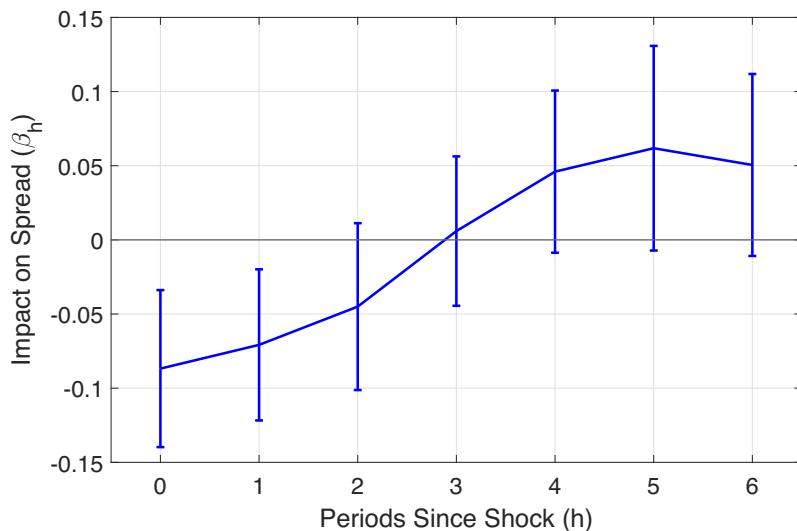


FIGURE III

## Projections of Spreads on Monetary Policy Shocks

The figure plots the point estimates and 95% confidence intervals for the  $\beta_h$  coefficients in [equation \(30\)](#). The coefficients are the projection estimates of sovereign spreads on monetary policy shocks.

try  $c$  at time  $t$  on a vector of monetary shocks with  $L$  lags and additional controls  $Z_{c,t}$ ,

$$(29) \quad \text{spr}_{c,t} = \sum_{h=0}^L \beta_h m_{c,t-h} + Z_{c,t} \Gamma + v_{c,t}.$$

The coefficients of interest are the  $\beta_h$ 's. We include in the controls  $Z_{c,t}$  contemporaneous values for output growth and inflation, as well as nine lags for these variables. We set  $L = 6$  for our baseline specification. The identifying assumption is that with these control variables, the coefficients  $\beta_h$  isolate the effects of monetary shocks on fluctuations in spreads. We also cluster standard errors across time and country.

[Figure III](#) plots the estimated coefficients  $\beta_h$  together with their 95% confidence band. Contractionary monetary policy shocks tend to decrease spreads. On impact, a positive monetary policy shock of one standard deviation decreases spreads by about 0.09 standard deviations. The effects are also persistent, as the point estimate is negative for three periods, but the significance

decreases. Given these estimates and that the mean standard deviation of monetary shocks is 0.23 and that for spreads is 0.9, the mean estimate for the first three months of  $-0.07$  implies an elasticity of spreads to monetary shocks of  $-0.3$ , with confidence bands reaching  $-0.6$  in the first month. In [Online Appendix C.2](#), we provide more details on our exercise and document the robustness of these results.

#### IV.B. Functional Forms and Parameterization

We assume preferences are separable between a composite consumption good  $H$  and labor  $N$ . The per period utility function is given by

$$u(C_t, C_t^f, N_t) = \log[H(C_t, C_t^f)] - \frac{N_t^{1+\frac{1}{\xi}}}{1+\frac{1}{\xi}},$$

where  $H(C_t, C_t^f)$  is a CES composite of domestic goods  $C$  and imported goods  $C^f$ ,

$$H(C_t, C_t^f) = \left( \theta C_t^{\frac{\omega-1}{\omega}} + (1-\theta)(C_t^f)^{\frac{\omega-1}{\omega}} \right)^{\frac{\omega}{\omega-1}}.$$

We can derive the CPI as the price of the bundle of domestic- and foreign-goods consumption,  $P^{\text{CPI}} = P^d [\theta^\omega + (1-\theta)^\omega e^{1-\omega}]^{\frac{1}{1-\omega}}$ , and the resulting CPI inflation,  $\pi^{\text{CPI}} = \frac{P^{\text{CPI}}}{P_{-1}^{\text{CPI}}}$ , where the subscript  $-1$  denotes the previous period's value. The rate of depreciation of the nominal exchange rate is given by

$$(30) \quad \frac{\varepsilon}{\varepsilon_{-1}} = \frac{e}{e_{-1}} \frac{P^d}{P_{-1}^d} = \frac{e}{e_{-1}} \pi,$$

which reflects inflation and the depreciation of the terms of trade.

The model features productivity, monetary, and enforcement shocks. We assume that the productivity and monetary shocks follow independent AR(1) processes,  $\log \tilde{z}_t = \rho_z \log \tilde{z}_{t-1} + \sigma_z \varepsilon_{z,t}$  and  $\log m_t = \rho_m \log m_{t-1} + \sigma_m \varepsilon_{m,t}$ , with innovations  $[\varepsilon_{z,t}, \varepsilon_{m,t}] \sim \mathcal{N}(0, I_2)$ . The productivity costs from default follow [Chatterjee and Eyigungor \(2012\)](#) such that  $z(\tilde{z}, \Theta) = \tilde{z} - \max\{0, \lambda_0 \tilde{z} + \lambda_1 \tilde{z}^2\}$  if  $\Theta = 1$ , and  $z(\tilde{z}, \Theta) = \tilde{z}$  if  $\Theta = 0$ . The enforcement shocks  $v$  are i.i.d. and perturb the value of default. We integrate these shocks into our computational technique following [Dvorkin et al. \(2021\)](#) and

Mihalache (2020) such that  $v$  follows a logistic distribution, where the parameter  $\varrho_D$  controls the relative importance of the shocks.<sup>24</sup>

We consider a quarterly model and set some parameters externally, based on prior studies, whereas others are determined internally as part of a moment-matching exercise. We first describe the parameters set based on direct measurement or previous work. In terms of preference parameters, we set the elasticity of substitution between foreign and domestic goods in consumption  $\omega$  to 0.85 following Corsetti, Dedola, and Leduc (2008), the weight of domestic goods in consumption  $\theta$  is set to induce an imports' share in the balanced-trade steady state of 29% based on our emerging-market data, and the elasticity of labor supply  $\zeta$  is set to  $\frac{1}{3}$  as in Galí and Monacelli (2005). In terms of technology parameters, we set the domestic varieties' elasticity  $\eta$  to 6, which corresponds to a 20% markup, in line with the estimates in Edmond, Midrigan, and Xu (2023) and Díez, Fan, and Villegas-Sánchez (2021), the Rotemberg adjustment cost  $\varphi$  is set such that the frequency of price changes is roughly once a year, using the well-known equivalence between Calvo and Rotemberg pricing,<sup>25</sup> and the persistence of the productivity shock is set to 0.9 based on international business cycle studies. Other parameters we set are the export demand elasticity  $\rho$  to 5 based on Bajzik et al. (2020); the international interest rate  $r$  to 0.5 implying a 2% annual rate, consistent with U.S. Treasury yields; the probability of return to financial markets after default  $\iota$  such that the average length of market exclusion is six years, based on Cruces and Trebesch (2013); the debt-decay parameters  $\delta$  such that the duration of debt is six years, consistent with emerging-markets data; the persistence and standard deviation of monetary shocks to 0.3 and 0.23%, respectively, based on our estimates of monetary shocks; and the intercept of the interest rate rule to satisfy the steady-state condition  $\bar{i} = \frac{\pi}{\beta}$ . Finally, we normalize the level of export demand  $\xi$  to one.

The second set of parameters is pinned down by a moment-matching exercise, such that our model replicates salient features of emerging-market inflation targeters' data. These eight param-

24. This computational technique consists of augmenting the model with taste shocks in the discrete-choice tradition. Online Appendix G contains more details on the computational algorithm and simulation of the model.

25. See Miao and Ngo (2021) for the mapping between the Calvo and Rotemberg parameters.

eters are the discount factors of the private sector and that of the government,  $\beta$  and  $\beta_g$ , respectively, the inflation target  $\bar{\pi}$ , the interest rate rule coefficient  $\alpha_P$ , the volatility of the productivity innovations  $\sigma_z$ , the parameters of the default cost function  $\{\lambda_0, \lambda_1\}$ , and the parameter governing the magnitude of the enforcement shock  $\varrho_D$ . We target nine average moments in the data: averages of the means and standard deviations of CPI inflation and spreads, means of nominal domestic rate, standard deviations of output and consumption, the mean ratio of government debt to output, and the correlation of spread with output.<sup>26</sup>

Most parameters affect all moments, yet some moments are more informative for setting certain parameters. The average CPI inflation rate in the data is the most influential for  $\bar{\pi}$ . The weight of inflation in the interest rate rule  $\alpha_P$  heavily affects the volatility of inflation. The volatility of productivity shocks is the main driver of output volatility. The enforcement-shock parameter  $\varrho_D$  and the productivity default cost parameters are crucial for the dynamics of spreads. The volatility of the enforcement shock informs the mean and standard deviation of spreads, while the productivity-cost parameters shape the correlation of spread with output and the debt level. The discount factor of the sovereign  $\beta_g$  affects the volatility of consumption and the level of debt, and the discount factor of the private sector  $\beta$  controls the average real domestic rate. We collect the values of all the parameters in [Table II](#).

We note that the parameters of the interest rate rule,  $\alpha_P$  and  $\bar{\pi}$ , are in line with external estimates. The value of  $\alpha_P$  is well within the range of estimates in the reaction functions we recovered for the eight emerging markets. Moreover, the resulting average rate of annual inflation, in our quarterly model, given by  $\bar{\pi}^4 \approx 1.04$ , is in line with the announced inflation targets in the countries of our sample. For example, the Central Bank of Brazil pursued a 4.5% target before 2018 and 3% in recent years, while South Africa's target has been 4.5% since 2017. The targets for Colombia and Chile were at 3% since the mid-2000s.

[Table III](#) reports the fit of the moment-matching exercise. Overall, we find that the model is able to replicate closely the targeted moments in the data. We compute model moments using a long simulated sample, from which we exclude default episodes and their associated international market exclusion spells be-

26. We focus on CPI inflation because of data limitations with alternative price indices. See additional details about the data in [Online Appendix C.1](#).

TABLE II  
PARAMETER VALUES

| Assigned parameters      |                                              | Parameters from moment matching |                                      |
|--------------------------|----------------------------------------------|---------------------------------|--------------------------------------|
| Preferences              | $\omega = 0.85, \theta = 0.73, \zeta = 0.33$ | Discount factor                 | $\beta = 0.9947$                     |
| Varieties elasticity     | $\eta = 6$                                   | Interest rate rule              | $\alpha_P = 1.575$                   |
| Export demand elasticity | $\rho = 5$                                   | Inflation target                | $\bar{\pi} = 1.010$                  |
| Price adjustment cost    | $\varphi = 58$                               | Government discount factor      | $\beta_g = 0.9775$                   |
| Productivity persistence | $\rho_z = 0.9$                               | Productivity volatility         | $\sigma_z = 1.3\%$                   |
| Monetary shock           | $\rho_m = 0.3, \sigma_m = 0.06\%$            | Default costs                   | $\lambda_0 = -0.45, \lambda_1 = 0.5$ |
| International rate       | $r = 0.5\%$                                  | Enforcement shock               | $q_D = 1.8e^{-4}$                    |
| Reentry probability      | $\iota = 4.17\%$                             |                                 |                                      |
| Debt duration            | $\delta = 0.037$                             |                                 |                                      |

TABLE III  
TARGETED MOMENTS

|                       | Data | NK-Default |
|-----------------------|------|------------|
| Means                 |      |            |
| CPI inflation         | 3.9  | 3.9        |
| Nominal domestic rate | 5.7  | 5.7        |
| Spread                | 2.0  | 2.0        |
| Debt to output        | 16   | 17         |
| Standard deviations   |      |            |
| Output                | 2.3  | 2.2        |
| CPI inflation         | 1.8  | 1.8        |
| Spread                | 0.9  | 0.9        |
| Consumption aggregate | 2.4  | 2.6        |
| Correlation           |      |            |
| (Output, spread)      | -42  | -53        |

cause none of the countries in our data sample outright defaulted during the sample period.<sup>27</sup> CPI inflation, nominal domestic rates, and spreads are reported annualized. In the model and data, CPI inflation is about 3.9%, spreads are 2%, nominal domestic rates are 5.7%, and debt is about 16% of output. The model also replicates closely the standard deviation of output, inflation, and consumption. The volatility of spreads is 0.9% in both the data and the model. Output is negatively correlated with spreads, with similar correlations in the data and the model.

#### IV.C. Policy Rules and Impulse Response Functions

To showcase the workings of the quantitative model, we describe policy functions and plot impulse response functions following monetary and productivity shocks. These enable us to illustrate how our mechanisms, default amplification and monetary discipline, are operational in the quantitative model.

1. *Policy Rules.* [Figure IV](#) presents policy rules as a function of debt  $B$ . To highlight the impact of default risk on policy functions, we plot two lines in each figure, a solid line for the variable of interest and a dotted line for the equilibrium, one-period-ahead default risk, using a second, right-side vertical axis. In the figures,  $B$  is scaled by average annual output in foreign goods and

27. In the model the default frequency is modest and equal to 1.75%, which resembles the average spread given risk-neutral pricing.

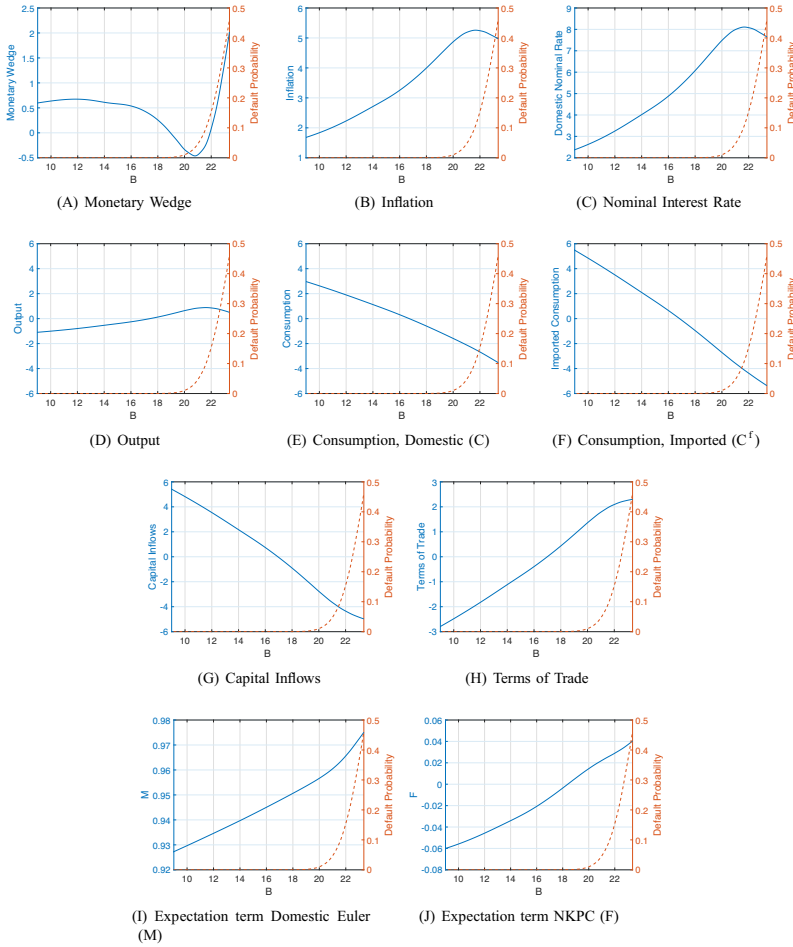


FIGURE IV  
Equilibrium Policy Functions

Inflation and the nominal interest rate as expressed in percentage points. Output, domestic consumption, imported consumption, and the terms of trade are expressed as log point deviations from the level at the lowest  $B$  on the horizontal axes. Capital inflows and the debt level are scaled by average annual output in foreign goods. Quantities in Panels D–F share a common vertical axis range, to ease comparison.

the functions are for the average values of the productivity and monetary shocks, with the focus on behavior conditional on not defaulting. As shown by the dotted lines in the plots, default probabilities increase with current debt  $B$  because debt due next period  $B' = H_B(s, B)$  increases with  $B$ , which in turn makes a future default more likely.

Default risk affects domestic allocations in our model through two main channels. First, through the expectation channel: default risk alters the expectation terms of the domestic Euler equation and the NKPC. This channel was the focus of the theoretical analysis in [Section III](#). In that section, under a simplified set-up with a utility function linear in imported consumption  $C^f$ , default risk increased the expected marginal utility because of the productivity costs from default. In our quantitative model, preferences are concave and non-separable in imported consumption, which means expected marginal utility depends on future credit restrictions reflected in the bond-price function. High debt due next period is associated with elevated default risk, which pushes up expected marginal utility, even absent an actual default, because it is expensive to roll over the high debt that the economy inherits. These effects on marginal utility tend to increase the expectation terms of the domestic Euler equation and the NKPC, namely,  $F$  and  $M$  from [equations \(22\)](#) and [\(23\)](#). Panels I and J illustrate this expectation channel: both  $F$  and  $M$  functions increase with debt.

Second, in our general model, default risk also affects allocations through a terms-of-trade channel. Debt and default risk now affect the terms of trade contemporaneously through their impact on capital flows and imported consumption. To highlight the two channels, it is useful to describe the policy rules in [Figure IV](#), by drawing a distinction between two regions: a region with positive default risk for levels of  $B$  above 20%, where the expectations channel is mostly at play, and another one with essentially no default risk, for lower debt levels, where the terms of trade channel dominates.

Panels A and B illustrate key outcomes of our general model, which align with the theoretical predictions from [Proposition 1](#). Both the monetary wedge and inflation tend to be high with high default risk and high debt. The monetary wedge increases with  $B$  when default probabilities are positive. High default risk increases future expected marginal utility, and through the domestic Euler equation, it lowers current domestic consumption, as seen

in Panel E. Low demand for domestic consumption lowers output as well, as seen in Panel D in the positive–default risk region.

According to the NKPC [equation \(8\)](#), inflation is negatively related to the monetary wedge and positively related to the value of expected future inflation, encoded in  $F$ . The value of expected inflation is always increasing in debt—as shown in Panel J—and drives the overall inflation policy. In the low–default risk region, increases in debt lead to a depreciation of the terms of trade, which tends to increase employment and inflation. In response to high inflation, the monetary authority’s interest rate rule calls for higher nominal rates, as captured in Panel C.

Panel G plots the equilibrium capital inflows, given by  $q(s, B')(B' - (1 - \delta)B) - (r + \delta)B$  in our model with long-term debt. We express these inflows relative to average annual output in foreign goods. As default risk is reflected in the bond-price schedule, government borrowing increases slower than current debt, and hence capital inflows decrease as a function of debt  $B$ , resulting in a decline in imported consumption  $C^f$ , as shown in Panel F. In the region of low debt, with no default risk, this effect leads to a depreciation of the terms of trade (Panel H), which boosts export demand and also modestly increases output.

Note that the patterns in the low-default region mirror results in standard models in the literature. As [Blanchard et al. \(2017\)](#) discuss, in the workhorse open-macro model, a reduction in capital inflows is expansionary because it depreciates the exchange rate, the so-called expenditure switching effect. The novel finding in our model with default risk is that a reduction in capital inflows can be recessionary, as exemplified in the high-default region. This is because a reduction in capital inflows due to high default risk leads to a reduction in domestic consumption demand through the expectation channel, which in turn depresses production.<sup>28</sup>

*2. Impulse Response Functions.* We can further explore the mechanisms driving our model by plotting impulse responses for the main variables, following monetary and productivity shocks. We construct the impulse response functions (IRFs) in our nonlinear model following [Koop, Pesaran, and Potter \(1996\)](#). We simulate a panel of 1,500,000 units for 1,500 periods. For the first 1,450 pe-

28. Alternative financial frictions, such as the collateral constraint of [Mendoza \(2010\)](#), can also generate output contractions from reductions in capital flows.

riods, the shocks follow their underlying Markov chains so that the cross-sectional distribution converges to the ergodic distribution of the model. In period 1,451, the impact period (normalized to zero in the plots), we shock all units of the panel by the same amount. From period 1,452 onward, shocks again follow their Markov processes. The impulse responses plot over time the cross-sectional average taken over units of the panel. The impulse responses are computed over all units, including those with defaults. Discarding defaults from the cross-sectional average does not alter the salient properties of the IRFs.

We start with IRFs for a contractionary monetary shock to illustrate the disciplining force in our model, namely, that a high nominal interest rate lowers default risk. An increase in the monetary shock leads to dynamics that are standard in the New Keynesian literature: temporary increases in the nominal rate and temporary decreases in inflation, output, and consumption. In our NK-Default model, monetary shocks also affect the government's borrowing incentive. Contractionary monetary shocks encourage a reduction in borrowing, which reduces sovereign default risk and spreads, in line with the discipline mechanism.<sup>29</sup> The government reduces borrowing in response to contractionary shocks because doing so alleviates the monetary wedge induced by tight monetary policy. The responses in [Figure V](#), Panels A–F imply that a contractionary monetary shock of about 3% increases nominal rates by 2, decreases inflation by about 0.7, and decreases spreads by about 0.1. The elasticities for nominal rates and inflation are within the bounds recently estimated by [Ha et al. \(2025\)](#) using data from emerging markets. The average spread response masks a large heterogeneity of the effect across units (in the limiting distribution), where it reaches a decrease of over 1.9%. This range of elasticities is consistent with our empirical findings and the inflation event analysis below, where we find a peak elasticity of  $-0.33$ . Output and consumption drop by about 0.2 and 0.7, respectively.

[Figure VI](#) collects the IRFs with respect to the productivity shock. Output declines about 3.5%, with a contractionary productivity shock of comparable magnitude. Consumption declines somewhat more because, as is typical in sovereign-default models, low productivity tightens the bond-price schedule due to elevated default risk. The tight bond-price schedule leads to higher

29. [Online Appendix Figure A.3](#) shows that default risk decreases with monetary shocks, regardless of the current level of debt.

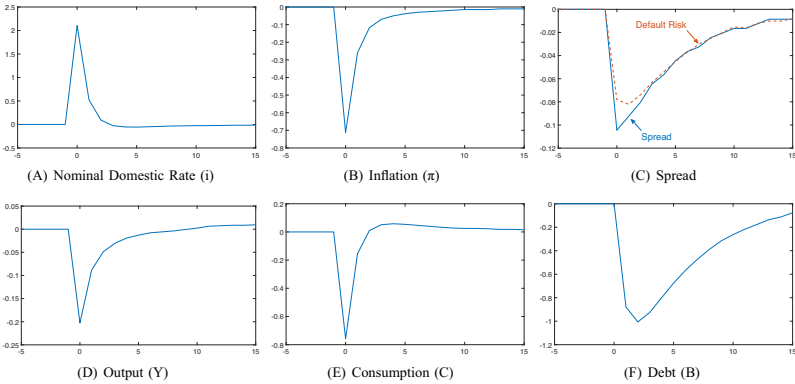


FIGURE V  
Impulse Response Functions to Monetary Shocks

Impulse response functions to a contractionary monetary shock  $m$ . All variables are expressed as percentage deviations from their sample average, except for the spread and default risk, which are level differences.

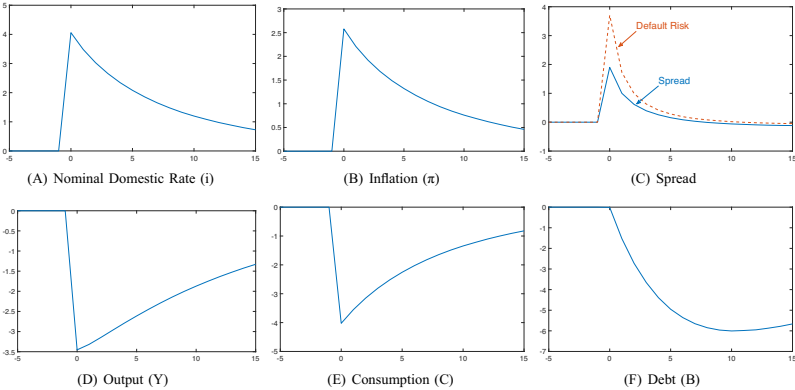


FIGURE VI  
Impulse Response Functions to Productivity Shocks

Impulse response functions to a contractionary productivity shock,  $z$ . The panels plot variables expressed as a percentage deviation from the sample average, except for the spread and default risk, which are level differences.

spreads and a reduction in debt, which tends to increase the volatility of consumption. Spreads rise about 2%, default risk rises about 3.7%, and debt contracts slowly, by about 6% of its average value. Spreads rise by less than default risk because the spread

is the average default risk for the duration of the bond. Inflation rises about 2.5% on impact because of the high unit cost from low productivity and the increased default risk. The nominal domestic rate increases in response to the elevated inflation, about 4%. These dynamics illustrate that productivity shocks lead to a strong, positive comovement of spreads with inflation and nominal rates.

#### *IV.D. Business Cycle Moments and the Reference Model*

We turn to our model's business cycle implications and compare them with the moments of a reference model without default risk. We show that our model can deliver key untargeted patterns in the data and that default risk amplifies the volatility of inflation. We start by introducing a reference model without default.

*1. NK-Reference.* To isolate the interactions between monetary frictions and default risk, we compare our findings with a reference model, labeled NK-Reference, which is a version of the [Galí and Monacelli \(2005\)](#) model with nominal rigidities and without default. The equilibrium of the NK-Reference model is characterized by the same private and monetary equilibrium of our baseline NK-Default model from [Definition 1](#) and summarized in the [Online Appendix](#) by equations (32)-(37), the international Euler condition (18), and an exogenous debt-elastic bond-price schedule to close the model, as in [Schmitt-Grohé and Uribe \(2003\)](#). The debt-elastic bond-price schedule is  $q^{\text{ref}}(B)^{-1} = \beta + \Gamma[\exp(B - \bar{B}) - 1]$ , with  $\Gamma$  set to  $10^{-5}$ , which gives a very loose borrowing schedule, and  $\bar{B}$  set to give the same average debt level as our baseline. We solve the NK-Reference model with a first-order log-linear approximation of the equilibrium conditions, keeping all parameter values as in the baseline.

*2. Business Cycle Statistics.* The first two columns of [Table IV](#) report the first and second moments for the emerging-market data and for simulated data from our baseline NK-Default model. The data consist of the series reported in [Sections IV.A](#) and [IV.B](#), for output, CPI inflation, nominal domestic rates, spreads, government debt to output, and aggregate consumption, as well as the trade-weighted nominal exchange rate depreciation. All moments are averages across the emerging markets in our sample and are reported in percentage points.

TABLE IV  
COMPARISON OF MOMENTS ACROSS DATA AND MODELS

|                             | Data | NK-Default | NK-Reference | NK-Default alternative rules |                   |
|-----------------------------|------|------------|--------------|------------------------------|-------------------|
|                             |      |            |              | Strict inflation             | Inflation default |
| Mean                        |      |            |              |                              |                   |
| CPI inflation               | 3.9  | 3.9        | 4.1          | 4.3                          | 3.4               |
| Nominal domestic rate       | 5.7  | 5.7        | 6.2          | 5.9                          | 5.3               |
| Spread                      | 2.0  | 2.0        | —            | 2.5                          | 0.3               |
| Debt                        | 16   | 17         | 17           | 18                           | 20                |
| Standard deviation          |      |            |              |                              |                   |
| Output                      | 2.3  | 2.2        | 3.3          | 2.3                          | 2.3               |
| CPI inflation               | 1.8  | 1.8        | 1.0          | 0.2                          | 0.4               |
| Spread                      | 0.9  | 0.9        | —            | 0.7                          | 0.1               |
| Consumption aggregate       | 2.4  | 2.6        | 1.6          | 3.3                          | 3.7               |
| Nominal domestic rate       | 1.9  | 3.1        | 1.6          | 2.3                          | 3.8               |
| Depreciation rate           | 8.6  | 1.9        | 1.6          | 1.0                          | 1.1               |
| Correlations                |      |            |              |                              |                   |
| (Spread, output)            | −42  | −53        | —            | −63                          | −59               |
| (Spread, CPI inflation)     | 46   | 59         | —            | −13                          | 59                |
| (Spread, nom. dom. rate)    | 30   | 78         | —            | 56                           | 93                |
| (Spread, depreciation rate) | 37   | 52         | —            | −10                          | 25                |
| (CPI inf., output)          | −22  | −15        | −90          | 7                            | −10               |
| (CPI inf., nom. dom. rate)  | 59   | 90         | 90           | −3                           | 56                |
| Autocorrelations (%)        |      |            |              |                              |                   |
| Output                      | 84   | 69         | 94           | 71                           | 74                |
| CPI inflation               | 88   | 96         | 91           | 72                           | 59                |
| Spread                      | 83   | 63         | —            | 51                           | 71                |
| Nominal domestic rate       | 95   | 85         | 90           | 60                           | 49                |

Notes. This table presents business cycle moments for the data and four models: the baseline NK-Default model, the reference model without default risk NK-Reference, and two alternative monetary policy regimes, strict inflation targeting and the inflation–default rule of [Section IV.F](#), which also targets default risk. [Online Appendix C.1](#) details the construction of our sample and the data moments.

Overall, the moments in the baseline model resemble those of emerging-market data. The mean CPI inflation, nominal domestic rate, spread, and debt, as well as the volatility of inflation, output, and spreads are targets in our moment-matching exercise. The model delivers volatility of the nominal rate comparable to the data, whereas it underestimates the high volatility in nominal exchange rates, reflecting the common disconnect between exchange rates and fundamentals in much of international business cycle theory. Our model mirrors the cyclical properties of the trade bal-

ance, with a correlation between the trade-balance-to-output ratio and output of  $-24\%$  in the data versus  $-42\%$  in the model.

The model delivers the key stylized fact that inflation correlates positively with spreads, close to  $50\%$  in the model and the data. In addition, the model generates a positive correlation between the nominal rate and the nominal depreciation rate with spreads. These correlations arise in our model because, across all state variables, namely, the productivity shock  $z$ , monetary shock  $m$ , and debt  $B$ , inflation and spreads comove positively. These positive correlations are the implications of the amplification and disciplining mechanisms of our model. Finally, the model also delivers autocorrelations for output, inflation, spreads, and nominal domestic rates in line with those in the data.

To provide further evidence of the model's default amplification mechanism, which relates default risk to elevated inflation expectations, we assemble surveys of inflation expectations for several countries in our sample. As documented in [Online Appendix C.3](#), we find that in the data, a  $1\%$  increase in spreads is associated with an average  $0.97\%$  increase in inflation expectations for the following 12 months. The corresponding measure, in our baseline NK-Default model, is  $1.14\%$ , well within the  $95\%$  confidence interval of the estimated coefficient.

The third column of [Table IV](#) reports the moments of the NK-Reference model, which is silent on default risk. Average CPI inflation and the nominal domestic rate are similar to the benchmark. The standard deviations of CPI inflation and the nominal interest rate are, however, only about  $55\%$  of those in the NK-Default baseline. Default risk makes inflation more volatile because it affects expected future inflation and the monetary wedge, as illustrated in [Figure IV](#).<sup>30</sup> In [Online Appendix H](#) we also illustrate these amplification effects from default risk with IRFs. In the NK-Reference model, the response of inflation to productivity shocks is about half of the response in the baseline with default. In addition, monetary policy is more effective in reducing inflation; the inflation response to contractionary monetary policy is more than double in the NK-Reference, relative to the baseline. This comparison shows that an emerging-market central bank targeting

30. In this NK-Reference model, monetary wedge and inflation policies are extremely flat as a function of debt because of flat expectations encoded in functions  $F$  and  $M$  in [equations \(22\)](#) and [\(23\)](#), unlike the functions in our baseline model.

inflation must implement a more volatile interest rate policy due to sovereign default risk.

#### *IV.E. Temporary Inflation Events*

We revisit the temporary inflation events. We use our baseline model to evaluate its quantitative performance in matching the observed dynamics of the events discussed in [Section IV.A](#). We perform two counterfactuals to assess the roles of default risk and monetary policy during such episodes, by comparing our baseline to an economy without default and one with tighter monetary policy.

To simulate the event, we start the model at the mean of the stationary distribution, with good credit standing, and feed in sequences of productivity and monetary shocks. The shocks are chosen such that the resulting paths of inflation and the nominal interest rate in our NK-Default baseline model best fit the data in [Figure II](#). The solid blue lines in [Figure VII](#) represent the resulting model time paths for inflation, nominal rates, output, and spreads. The black diamond markers correspond to the data. All series are standardized, as described in [Section IV.A](#). The baseline model replicates well the paths for inflation and the nominal rate, as seen in the top two panels. Note that in our nonlinear model with default risk, there is no guarantee that there exist shock paths that can deliver arbitrary paths for these variables, so we view this result as providing further validation of our model. Inflation increases temporarily, close to 2.5 standard deviations, in both model and data, and returns to low levels about a year after peaking. Nominal rates increase by 1.5 standard deviations as inflation picks up, and then they fall below their long-term average once inflation subsides. As shown in [Online Appendix Figure A.4](#), inflation events result from the interaction of low productivity and expansionary monetary policy shocks.

The baseline model delivers comparable dynamics to the data for output and spreads, as illustrated in the bottom two panels of [Figure VII](#). Output falls to about two standard deviations below average and the spread increases by about three standard deviations, quickly returning to average levels within six quarters from the peak of inflation. In the data, nevertheless, the trough in output is about two quarters after the trough in the model. Given our model's account of these events, we use it to shed light on the role of default risk and monetary policy through two counterfactuals.

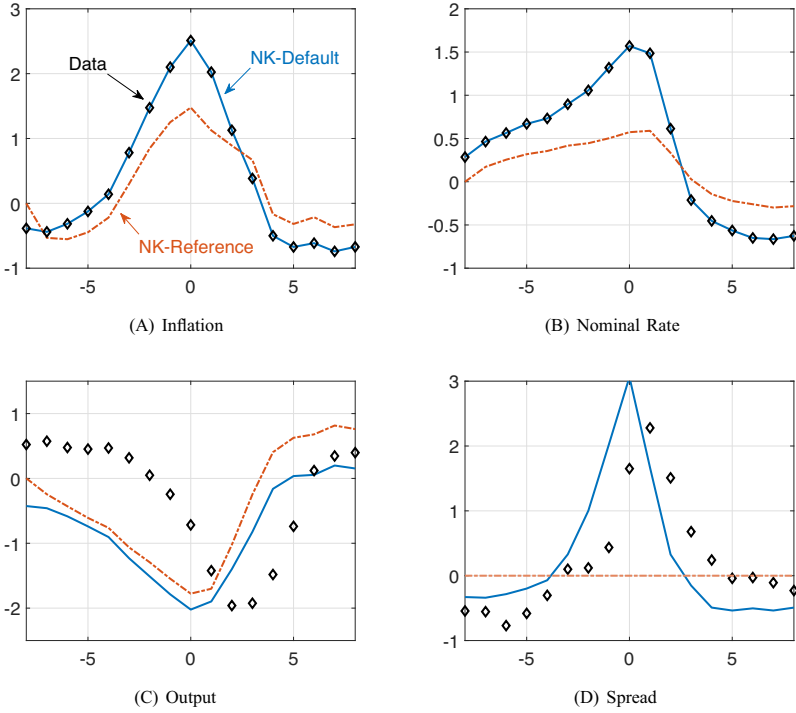


FIGURE VII

## Inflation Event Analysis: Model Fit and the Role of Default Risk

Time paths for productivity  $z$  and monetary shocks  $m$  are set to fit the paths of inflation and the nominal rate in the baseline model. Shock paths are plotted in [Online Appendix Figure A.4](#). The data are in black diamond markers, the same quarterly paths as in [Figure II](#). The blue solid line is the baseline model and the dashed red line is the NK-Reference model without default.

For our first counterfactual, we assess the contribution of default risk to these dynamics by comparing our baseline NK-Default model to paths produced by the NK-Reference model without default risk. We feed the NK-Reference model the same time paths for productivity and monetary shocks, and start the episode in the steady state. The resulting series are the dashed red lines in [Figure VII](#). Without default risk, the increase in inflation during the event is more muted than in the baseline, about 50%, from the start of the event to the peak. In contrast, the decrease in output in the NK-Reference model is comparable to that of the baseline model with default. Panel B shows that the increase in nominal

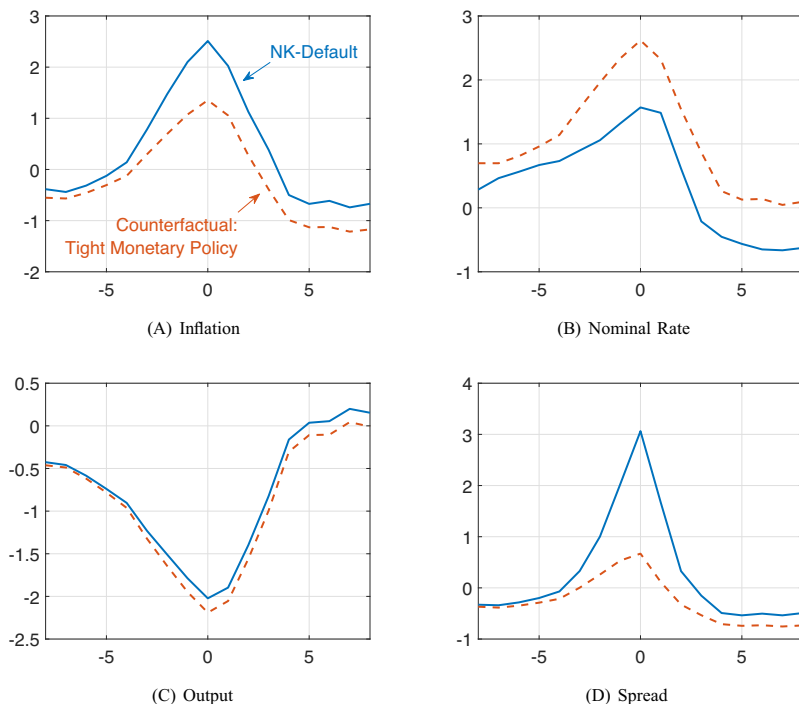


FIGURE VIII

## Inflation Event Analysis: The Role of Monetary Policy

See the note to [Figure VII](#). The solid blue line is the baseline model while the dashed red line is the counterfactual with tight monetary policy. Time paths for productivity  $z$  and monetary shocks  $m$  are plotted in [Online Appendix Figure A.4](#).

rates is smaller, reflecting the more muted rise in inflation. Finally, note that the NK-Reference model is silent on spread dynamics.

For a second counterfactual, we evaluate the role played by monetary policy in shaping temporary inflation events. We compare our baseline-model time paths with a case in which monetary shocks are contractionary throughout, as presented in [Online Appendix Figure A.4](#). As shown in [Figure VIII](#), these shocks deliver a one standard deviation increase in the nominal rate at its peak, a 35% smaller increase in CPI inflation, and a slightly deeper recession, from the start of the event to the peak of inflation. Notably, the rise in spreads is much more modest, a reduction by about 70% compared to the baseline, over the same horizon. This exercise implies that at the peak of the event, a 1% increase in

the monetary shock decreases spreads by 0.33%. Interestingly, the elasticity of spreads to monetary shocks is heterogeneous across the state space and ranges between  $-0.02$  and  $-0.33$  in this event counterfactual.

The first counterfactual shows that during the temporary inflation events default risk is an important driver of inflation. Low productivity and expansionary monetary shocks tend to raise default risk, which increases expected inflation. In turn, this higher expected inflation feeds into current inflation via the firms' pricing decisions and calls for more aggressive monetary policy in the form of a stronger response by nominal domestic rates. The message from this event analysis mirrors the findings from the business cycle comparisons in [Table IV](#): default risk amplifies inflation volatility. Our second counterfactual illustrates how contractionary monetary policy can have a large effect on spreads, in addition to its traditional effect on inflation and activity, but also points to the state-contingent nature of these effects. To explore further the effect of monetary policy on the dynamics of the economy, we turn to alternative monetary policy rules in the next section.

#### IV.F. *Alternative Monetary Rules*

We compare our baseline NK-Default model, with an inflation-targeting monetary rule, to two different monetary policy regimes. The first alternative we consider is that of strict inflation targeting. Under this regime, the monetary authority sets nominal rates such that inflation  $\pi_t$  is at all times at the target level  $\bar{\pi}$ . Recall that this would be the optimal monetary regime for a model similar to ours, except without default risk. As in the theoretical analysis of [Section III](#), under strict inflation targeting, outcomes are those that would prevail under flexible prices.

The second alternative regime we consider is one under which monetary policy follows an augmented interest rate rule. Based on our theoretical analysis, we know that interest rate rules that respond to default risk can support better outcomes, therefore in this extension we propose a rule that responds to both inflation and default risk  $\Phi_t$ . For the inflation–default risk regime, the interest rate rule is

$$(31) \quad i_t = \bar{i} \left( \frac{\pi_t}{\bar{\pi}} \right)^{\alpha_P} \left( \frac{\Phi_t}{\bar{\Phi}} \right)^{\alpha_D} m_t.$$

We focus on the case under which  $\alpha_P$  is kept as in the baseline and where we fix a value for  $\bar{\Phi}$  corresponding to a 0.25% annual

default probability and an  $\alpha_D$  of eight. In [Section VI](#), we vary both  $\alpha_P$  and  $\alpha_D$  and report the welfare consequences of these alternative parameter values.

*1. Business Cycle Moments.* [Table IV](#) reports the business cycle moments for the alternative monetary policy regimes. The second column corresponds to our NK-Default baseline model while the fourth and fifth columns correspond to the strict inflation targeting and the inflation–default risk regimes, respectively. Under strict inflation targeting, the mean spread is roughly 50 basis points higher than in NK-Default, while CPI inflation is substantially smoother. In this regime, the monetary discipline mechanism is disabled and, as a result, spreads are higher due to more aggressive sovereign borrowing. This model exhibits two counterfactual negative correlations, between spreads and CPI inflation and between spreads and the depreciation rate. As producer-prices inflation is constant at target, all movements in CPI inflation are due to changes in the terms of trade, and the correlation between spreads and CPI inflation reflects terms of trade movements alone.

Turning to the inflation–default risk regime, we find that spreads are significantly reduced. One notable consequence of this rule that targets default is the much lower volatility of CPI inflation, about one-fifth, even though the  $\alpha_P$  parameter is unchanged. This quantitative finding echoes [Proposition 3](#) from our theoretical analysis of monetary discipline, where the default-risk rule reduces default risk and the monetary wedge, associated with inefficient movements in inflation.

In our quantitative analysis, we also find that strict inflation targeting is dominated by both the baseline monetary rule and the inflation–default risk rule. Before we discuss these welfare results in detail in [Section VI](#), we move to establish the robustness of our mechanisms in several extension economies.

## V. ROBUSTNESS ANALYSIS AND MONETARY DISCRETION

We examine several extensions of our baseline model. We consider economies with local-currency debt, and alternative default costs, including monetary costs, a case without productivity costs of default, and domestic financial frictions. We show that as in the baseline model, default risk amplifies monetary distortions and leads to high inflation volatility across these extension models. We

also find that the disciplining effects of tight monetary policy are robust to these extensions. Finally, we use the local currency–debt economy to compare monetary rules to monetary policy under discretion.

#### *V.A. Local-Currency Debt and Monetary Discretion*

Our baseline NK-Default model features debt denominated in foreign currency. We turn to the case of local-currency government debt, with a valuation that fluctuates with inflation. This assumption is also empirically relevant because many emerging-market governments that are inflation targeters increasingly rely on domestic-currency debt for their financing.

We also use this economy to study the effects of discretionary monetary policy. As is well known, monetary policy with discretion in these environments involves a trade-off between using inflation to devalue debt and incurring costly pricing frictions.<sup>31</sup> Moreover, monetary policy is time inconsistent: *ex post*, the government has incentives to inflate away debt, but *ex ante* would have liked to commit to not inflate to receive better borrowing terms. In our economy with discretion, the government chooses local-currency borrowing, default, and nominal interest rates, period by period. The details of these models are found in [Online Appendix E](#).

The second and third columns in [Table V](#) report results for economies with local-currency debt under the baseline monetary rule (10) and under discretion, respectively. For these experiments, we keep the parameter values of the baseline calibration. Under the rule, mean inflation is less than half of the inflation under discretion (4.1% versus 9.3%). This is because under discretion the government actively uses monetary policy to inflate away the debt. The low volatility of inflation under discretion means that the government keeps inflation elevated across shock levels. In terms of sovereign spreads, the economy under discretion features an average spread that is more than double the spread in the monetary policy–rule economy.<sup>32</sup> The volatility of consumption is also much

31. We abstract from the potential benefits of discretionary monetary policy in the form of seigniorage revenue, a topic that [Espino et al. \(2025\)](#) explore in an environment with foreign-currency defaultable debt.

32. For comparability, we report sovereign spreads for synthetic foreign-denominated securities, priced in the economy with local-currency debt. In practice, however, local-currency spreads as defined by [Du and Schreger \(2016\)](#) have almost identical properties in our model.

TABLE V  
ROBUSTNESS ANALYSIS AND MONETARY DISCRETION

|                             | Local-currency debt |                |            | Alt. default costs |             |                   |
|-----------------------------|---------------------|----------------|------------|--------------------|-------------|-------------------|
|                             | NK-<br>Default      | NK-<br>Default | Discretion | Monetary           | No<br>prod. | Fin.<br>frictions |
| Mean                        |                     |                |            |                    |             |                   |
| CPI inflation               | 3.9                 | 4.1            | 9.3        | 3.9                | 4.2         | 3.8               |
| Nominal domestic rate       | 5.7                 | 6.0            | 10.8       | 5.8                | 6.1         | 5.6               |
| Spread                      | 2.0                 | 0.6            | 1.5        | 1.6                | 0.8         | 2.0               |
| Debt to output              | 17                  | 23             | 23         | 19                 | 18          | 18                |
| Standard deviation          |                     |                |            |                    |             |                   |
| Output                      | 2.2                 | 2.5            | 2.2        | 2.2                | 2.5         | 2.4               |
| CPI inflation               | 1.8                 | 1.7            | 0.4        | 1.9                | 1.7         | 1.8               |
| Spread                      | 0.9                 | 0.3            | 0.4        | 0.7                | 0.1         | 1.0               |
| Consumption aggregate       | 2.6                 | 2.3            | 3.3        | 2.7                | 2.3         | 2.7               |
| Nominal domestic rate       | 3.1                 | 2.8            | 2.2        | 3.2                | 2.8         | 3.1               |
| Depreciation rate           | 1.9                 | 1.8            | 0.8        | 1.9                | 1.9         | 2.0               |
| Correlations                |                     |                |            |                    |             |                   |
| (Spread, output)            | −53                 | −42            | −62        | −56                | −3          | −50               |
| (Spread, CPI inflation)     | 59                  | 71             | 9          | 56                 | 16          | 61                |
| (Spread, nom. dom. rate)    | 78                  | 90             | 65         | 77                 | 15          | 77                |
| (Spread, depreciation rate) | 52                  | 60             | 9          | 49                 | 7           | 51                |
| (CPI inf., output)          | −15                 | −3             | −17        | −17                | −1          | −12               |
| (CPI inf., nom. dom. rate)  | 90                  | 86             | −10        | 88                 | 88          | 87                |
| Autocorrelations (%)        |                     |                |            |                    |             |                   |
| Output                      | 69                  | 67             | 72         | 69                 | 69          | 68                |
| CPI inflation               | 96                  | 95             | 85         | 96                 | 98          | 96                |
| Spread                      | 63                  | 71             | 48         | 61                 | 85          | 65                |
| Nominal domestic rate       | 85                  | 83             | 55         | 84                 | 89          | 85                |

*Notes.* This table presents business cycle moments for six models. The first column repeats results for the baseline *NK-Default* model. In the second and third columns, government debt is denominated in local currency. For the *NK-Default* case, monetary policy follows the standard rule [equation \(10\)](#), while for the *Discretion* case monetary policy is chosen period by period by the government. Under the *Alt default costs, monetary* model, default changes the monetary rule to be more accommodative. Under *Alt default costs, no prod.*, default does not lead to a decline in productivity but instead reduces utility. Under *Alt default costs, fin. frictions*, firms use working capital loans.

more elevated under discretion. These results show that relative to discretion, monetary rules not only deliver better inflation outcomes but also lower spreads and lower volatility for consumption.

The economy with local-currency debt under the baseline monetary rule features similar dynamics in response to shocks. [Online Appendix Figures A.7](#) and [A.8](#) contain IRFs for the variables of interest for monetary and productivity shocks. As in the baseline, spreads fall with contractionary monetary shocks, al-

though the sensitivity is smaller in this economy. With domestic-currency debt, contractionary monetary policy also increases the real value of the debt due to lower inflation, counteracting the disciplining effects.

For reference, the first column of [Table V](#) lists the baseline calibration results. Recall that the local-currency NK-Default economy uses the same parameter values as the baseline, and thus any differences between these cases are exclusively due to the denomination of the debt. The patterns of inflation across these two economies are similar, but sovereign spreads are lower with local-currency debt. Higher inflation during recessions induces a hedging advantage for local-currency debt, useful to avoid borrowing at high interest rates and leading to lower spreads.

### *V.B. Alternative Costs of Default*

We turn to the robustness of our results with respect to the costs of default. We assess robustness because these costs play an important role in sovereign-default models, and causal empirical estimates are scarce.<sup>33</sup> First, we evaluate a monetary cost of default, namely, that the monetary rule becomes more accommodative during default. Second, we consider an economy where default leads to a cost in terms of utility, as opposed to the productivity penalty from the baseline. Third, we study an economy where default risk worsens credit conditions for firms. We find that our main mechanisms of amplification and discipline are robust to these alternative costs.

*1. Monetary Costs.* In our baseline model, monetary policy is always committed to its interest rate rule, even during defaults. As default is a time of fiscal distress, the monetary authority could be under pressure to abandon its commitment to target inflation. We consider here the case where monetary policy is loose after default. The monetary authority sets lower policy rates during default, such that  $i_t = (\bar{i} - \Delta)(\frac{\pi_t}{\bar{\pi}})^{\alpha_P} m_t$ , with  $\Delta > 0$  if  $\Theta = 1$  and zero otherwise. We set  $\Delta$  to 0.25%, which implies that nominal rates increase by about an extra 1% in default, and keep all other parameters as in the baseline.

33. One clean piece of evidence comes from [Hébert and Schreger \(2017\)](#), who use high-frequency data and find that higher default risk causes declines in stock market valuation.

The results for this case are reported in the fourth column of [Table V](#). Relative to the baseline, this economy features somewhat higher volatility of inflation and consumption and a lower mean and volatility of spreads. In this economy, the amplification mechanisms are stronger, raising the volatility of inflation and consumption. In response, the government borrows with less default risk to ameliorate the additional friction, resulting in lower and more stable spreads. [Online Appendix Figures A.9](#) and [A.10](#) report IRFs, which exhibit similar patterns to our baseline model. We conclude that our mechanisms, amplification and disciplining, are robust to environments when monetary policy is looser during defaults.

2. *No Productivity Costs.* In the baseline model, default leads to a reduction in productivity. To tease out the importance of this assumption, we consider instead a constant utility cost to defaulting. We assume that the enforcement shock  $\nu$  now has a positive mean  $\bar{\nu}$ , which we set to reproduce the same debt to output ratio as in the baseline. All other parameters are fixed at their baseline values.

The simulated moments for this economy are in the fifth column of [Table V](#). This economy generates moments for inflation and nominal rates comparable to the baseline NK-Default economy, but delivers very different moments for spreads and a lower volatility of consumption. The substantial inflation volatility of this model implies that the inflation amplification from default risk does not require productivity costs from default. Recall that inflation volatility, absent default (in the NK-Reference model) is much lower, at 1.0%. The spread, however, has a lower mean and volatility, is acyclical, and is only weakly correlated with inflation and the nominal rate. These spread differences lower the volatility for consumption. This failure to match spread dynamics is consistent with the findings in the sovereign-debt literature that productivity costs of default are important quantitatively. [Online Appendix Figures A.9](#) and [A.10](#) plot IRFs and document that our mechanisms are qualitatively robust to this alternative cost of default. The responses of inflation and nominal rates are comparable to the baseline, while the disciplining mechanism is more muted, as the spread reduction from contractionary monetary shocks is about half.

As in the baseline model, default risk here also brings inflation pressures by affecting allocations through the expectation channel, as it increases expected marginal utility and expected in-

flation. In fact, the functions  $M$  and  $F$  continue to be increasing in debt, as in the baseline model. In the simple example of [Section III](#), the productivity cost of default was the reason expected marginal utility increased, but in the general model expected tight bond prices coupled with high debt further reinforce this effect. Expected inflation increases because marginal utility is especially high in states of low productivity and high inflation.<sup>34</sup> The message from these results is that inflation amplification from default risk does not require productivity costs from default, but these costs are important quantitatively for the disciplining mechanism and crucial for the model to replicate the patterns in the data.

*3. Domestic Financial Frictions.* In our baseline model we abstracted from credit market frictions for firms, but in practice both monetary policy and default risk can affect domestic credit conditions.<sup>35</sup> For this robustness exercise, we incorporate a standard working capital friction for firms and assume that firms need to pay their wage bill by taking a working capital loan, with interest rate  $i_{ft}$ . This interest rate faced by firms depends on the nominal interest rate set by monetary policy and also on sovereign spreads and default, as these affect the balance sheet of financial intermediaries. In this set up, there are additional endogenous costs of default risk because it can trigger a domestic credit crunch. [Online Appendix F](#) describes this extension in detail. For calibration, we use the empirical findings from [Arellano, Bai, and Bocola \(2026\)](#) that measure a passthrough of 0.64 from sovereign spreads to firms' borrowing rates during sovereign-debt crises, and keep all other parameter values as in the baseline model.

The results for this case are in the sixth column of [Table V](#). The properties of the model with financial frictions are similar to the baseline. However, the mean and volatility of spreads are slightly higher because of the additional, endogenous distortion of production, especially when default risk is high. These additional spillovers from default risk also result in a slightly higher volatil-

34. We note that in this model, although high debt is associated with high inflation and high expected marginal utility, high debt is not associated with positive and large monetary wedges (as in the baseline, illustrated in [Figure IV](#)).

35. See [Christiano, Eichenbaum, and Evans \(2005\)](#) for the reference model of monetary policy's effect on working capital loans to firms and [Bocola \(2016\)](#) for domestic credit crunches from default risk, on which we build for this exercise.

ity to consumption. This model also delivers the positive correlations of spreads with inflation, nominal rates, and depreciation rates, and produces comparable IRFs for productivity and monetary shocks, as seen in [Online Appendix Figures A.9 and A.10](#).

## VI. WELFARE IMPLICATIONS

Finally, we evaluate the welfare implications of monetary policy rules and regimes. We consider welfare under different parameter values for the monetary rule (31), in the baseline NK-Default economy, as well as for the extension economies. We also study two other monetary policy regimes: strict inflation targeting and monetary discretion. A robust finding is that strict inflation targeting is not an optimal monetary policy in economies with default risk. Monetary rules that are sufficiently responsive to inflation or default risk provide disciplining benefits for sovereign debt.

### VI.A. *Baseline NK-Default Model*

We start with the welfare implications of alternative monetary policy regimes for the baseline model, as covered by [Table IV](#). We report the welfare of domestic households at a state given by a debt level of zero ( $B = 0$ ) and all shocks at their median levels ( $z = m = 1$ ), while the economy enjoys good credit standing.<sup>36</sup>

[Figure IX](#) compiles the findings. It plots welfare expressed as a consumption-equivalent percentage, by varying the parameters of the monetary rule (31), relative to the strict inflation-targeting regime. In the left panel, we vary  $\alpha_P$ , the coefficient controlling the response of domestic nominal rates to deviations of inflation from target, while keeping  $\alpha_D = 0$ . The red diamond marks the value from our baseline parameterization, which indicates that households marginally prefer the rule to strict inflation targeting. For low enough levels of  $\alpha_P$ , strict inflation targeting is preferable. However, when the monetary policy rule calls for more aggressive responses to inflation, the disciplining mechanism lowers default risk enough, so that a regime exhibiting some inefficient produc-

36. We compared welfare at a particular debt level because we want to hold constant the level of wealth across model economies. Moreover, the welfare ranking presented here persists across the entire state space, although welfare differences tend to decrease slowly with debt and flatten out for sufficiently high debt, but remain positive.

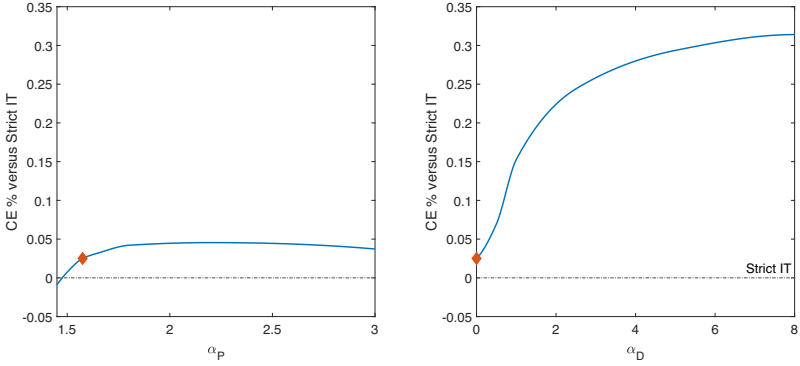


FIGURE IX

## Welfare Comparison Across Monetary Policy Regimes

Welfare gains of the monetary rule (31) expressed in consumption-equivalent measures, relative to the strict inflation-targeting regime, corresponding to the horizontal dashed black line at zero. The red diamond marks the level associated with our baseline NK-Default model. The left panel fixes  $\alpha_D = 0$  and varies  $\alpha_P$ . The right panel fixes  $\alpha_P = 1.575$  as in our baseline parameterization and varies  $\alpha_D$ .

tion due to pricing frictions, but where default risk is low, is better than outcomes under strict inflation targeting.

The right panel of Figure IX turns to our findings for the inflation–default risk monetary policy rule. Here,  $\alpha_P$  is kept at the baseline level and the horizontal axis varies  $\alpha_D$ , starting at its baseline value of zero. The welfare gains from adopting this rule over strict inflation targeting are potentially sizable and increase with  $\alpha_D$ . As  $\alpha_D$  approaches 8, households would be willing to forgo 0.3% of their consumption on average for policy to switch to the inflation–default risk rule instead of strict inflation targeting. We summarize these welfare results in the first four rows of the Table VI.

## VI.B. Model Extensions

We compile and summarize the welfare implications of different policy regimes, across the various model extensions. Table VI reports consumption-equivalent welfare, all relative to the baseline economy under a strict inflation-targeting policy. As before, the welfare statistics are computed for mean shocks and debt equal to zero. We also report the mean spread and inflation for each of the economies.

TABLE VI  
WELFARE, SPREADS, AND INFLATION ACROSS NK-DEFAULT ECONOMIES AND  
MONETARY POLICY RULES

|                                    | Welfare<br>(%) | Spread | Inflation |
|------------------------------------|----------------|--------|-----------|
| Baseline economy                   |                |        |           |
| Strict inflation target            | —              | 2.5    | 4.3       |
| Baseline monetary rule             | 0.025          | 2.0    | 3.9       |
| Less responsive monetary rule      | −0.009         | 2.1    | 3.8       |
| Inflation–default risk rule        | 0.314          | 0.3    | 3.4       |
| Local-currency debt                |                |        |           |
| Strict inflation target            | 0.081          | 1.9    | 4.3       |
| Baseline monetary rule             | 0.253          | 0.6    | 4.1       |
| Inflation–default risk rule        | 0.354          | 0.1    | 4.4       |
| Discretion                         | −0.195         | 1.5    | 9.3       |
| Financial frictions                |                |        |           |
| Strict inflation target            | 0.020          | 2.3    | 4.3       |
| Baseline monetary rule             | 0.025          | 2.0    | 3.8       |
| Inflation–default risk rule        | 0.342          | 0.2    | 5.4       |
| Monetary costs                     |                |        |           |
| Baseline monetary rule pre-default | 0.056          | 1.6    | 3.9       |
| No productivity costs              |                |        |           |
| Baseline monetary rule             | 0.326          | 0.8    | 4.2       |
| Shorter duration debt              |                |        |           |
| Strict inflation target            | 0.080          | 1.2    | 4.2       |
| Baseline monetary rule             | 0.084          | 0.7    | 4.0       |

*Notes:* This table reports welfare, mean spread, and mean CPI inflation across economies and monetary policy rules. Welfare is consumption equivalence evaluated for zero debt and the mean level of shocks. Welfare is reported relative to the baseline economy under a strict inflation-targeting monetary policy.

Table VI reports welfare for economies with local-currency debt, where firms face financial frictions, and with alternative default costs, which were discussed in Section V. We also compare welfare in the baseline economy, across sovereign-debt duration, by lowering it by two years, since maturity is a known determinant of the welfare losses from debt dilution (Hatchondo, Martinez, and Sosa-Padilla 2016). We evaluate several monetary regimes under each economy.

Consider welfare in the economy with local-currency debt. The baseline policy rule delivers sizable gains relative to strict inflation targeting (0.253% relative to 0.081%). These gains arise because the rule lowers spreads substantially and also because inflation patterns endow debt with a hedging advantage for the sovereign. The gains from the rule are double when compared to discretion (0.253% versus −0.195%). Discretion is the most detri-

mental monetary arrangement among all that we consider, because it generates high inflation and high spreads, relative to the case of local-currency debt and rule-based policy. The table also illustrates how issuing local-currency debt welfare-dominates issuing foreign-currency debt because of its aforementioned hedging properties.

For the economy with domestic financial frictions, the welfare ranking and magnitudes across monetary regimes are comparable to the baseline economy. These exercises are informative because they show that our welfare results in the baseline are robust to including endogenous costs of default from worsening domestic financial frictions.<sup>37</sup> Economies with monetary costs encode harsher costs from default, which leads to greater debt sustainability, lower spreads, and less default, all contributing to higher welfare. The case with no productivity costs features instead an additional punishment, in utils, making welfare not directly comparable to the baseline. However, this case exhibits very little default in equilibrium, which contributes to its higher household welfare.

Finally, the table also shows that the economy with shorter debt duration delivers higher welfare relative to the baseline, for a given monetary regime, consistent with prior work on sovereign-debt maturity. In addition, the monetary rule continues to be better than strict inflation targeting, but the gains are smaller than in the baseline case of longer duration, because of less debt dilution.

Comparing outcomes across these economies suggests that the fiscal and monetary regimes that deliver the best outcomes are those that result in fewer defaults and lower inflation. In [Figure X](#), we illustrate how economies with higher welfare tend to have lower mean spreads. The discretion economy features the lowest welfare level, even though its spreads are not exceptionally high. The low welfare of this economy is driven by its high average inflation rate. Interestingly, both the worst and the best policy regime features local-currency debt: the worst is discretionary policy with local-currency debt, while the best outcomes for households are achieved with local-currency debt and the inflation–default risk monetary policy rule.

37. See [Online Appendix Table A.4](#) for business cycle moments of this economy across monetary regimes.

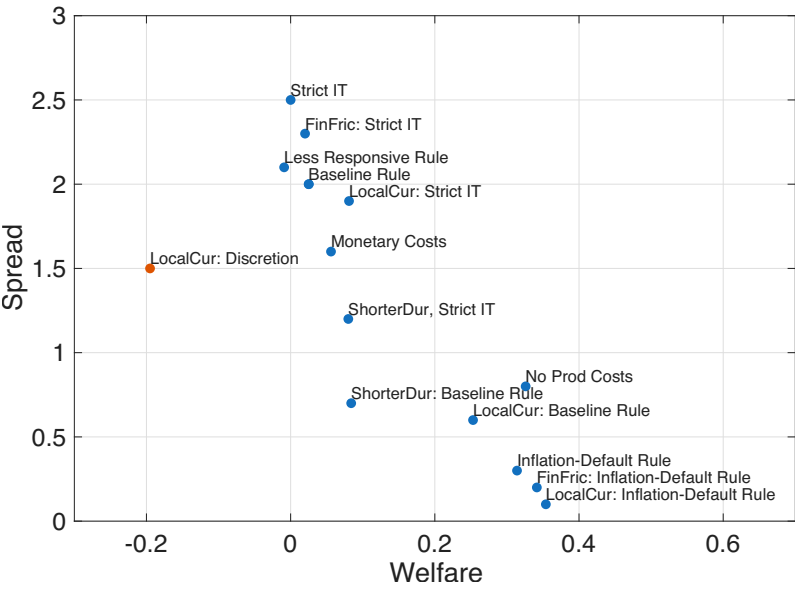


FIGURE X

Welfare and Spread Across Economies

Welfare is expressed as consumption equivalent, relative to the strict inflation-targeting regime in the baseline economy.

VII. CONCLUSION

We proposed a framework that combines two important aspects of current policy in emerging markets: sovereign risk in government debt and inflation targeting as monetary policy. It allowed us to identify novel mechanisms shaping the interplay of monetary policy, sovereign spreads, and domestic activity. We used our framework to study temporary inflation events in emerging markets, which have been accompanied by higher sovereign spreads and tighter monetary policy. We also provide support for the relevance of the mechanism of our model by combining empirical analysis using panel data of eight emerging-market inflation targeters with a quantitative evaluation of our model. Using our model as a building block, we established the robustness of our mechanisms to several extensions, including the currency denomination of the debt and domestic financial frictions for firms. Finally, our work speaks to the design of optimal monetary policy in emerging markets and finds that strict inflation targeting is not optimal in the presence of default risk.

In developing an integrated New Keynesian sovereign-default model, we have necessarily abstracted from important features affected by nominal rigidities and sovereign risk. Some of these include unemployment, investment, trade policies, the spillovers of U.S. monetary policy to emerging-market economies (Kalemlı-Özcan 2019), alternative assumptions about price stickiness such as local-currency pricing (Corsetti, Dedola, and Leduc 2010) or dominant-currency pricing (Gopinath et al. 2020), or other financial stability policies such as capital controls. We conjecture that some of the policy implications may be substantially altered when analyzed in a New Keynesian framework with sovereign risk.

Our framework could also be useful for the study of other episodes with elevated inflation or sovereign spreads. During the 2022 inflation outbreak, several European countries experienced increases in inflation and spreads jointly, similarly to the emerging-market episodes we studied. In contrast, during the 2012 European debt crisis the increases in spreads were not accompanied by inflation (De Ferra and Romei 2023). We leave for future work the question of whether a combination of supply and demand shocks can rationalize these distinct patterns within the context of our framework.

#### SUPPLEMENTARY MATERIAL

An Online Appendix for this article can be found at *The Quarterly Journal of Economics* online.

#### DATA AVAILABILITY

The data and code underlying this article are available in Arellano, Bai, and Mihalache (2026), in the Harvard Dataverse, <https://doi.org/10.7910/DVN/CMEFMV>.

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